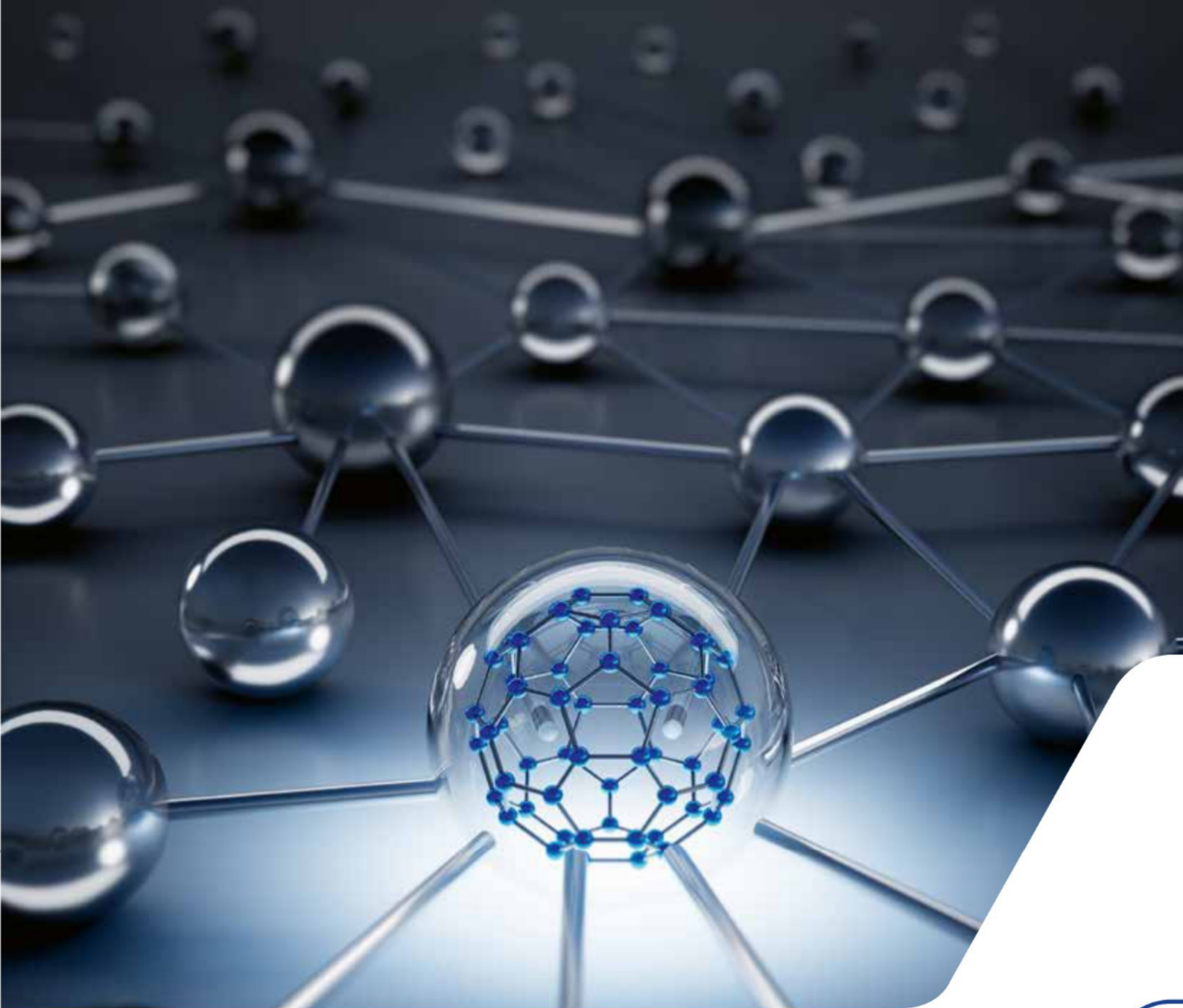




A2 Networks

What are the principles and concepts that underpin how networks operate?

SYLLABUS CONTENT

By the end of this chapter, you should be able to:

- ▶ A2.1.1 Describe the purpose and characteristics of networks
- ▶ A2.1.2 Describe the purpose, benefits and limitations of modern digital infrastructures
- ▶ A2.1.3 Describe the function of network devices
- ▶ A2.1.4 Describe the network protocols used for transport and application
- ▶ A2.1.5 Describe the function of the TCP/IP model (HL)

A2.1.1 The purpose and characteristics of networks

◆ **Computer network:** a system that connects computers and other devices to share resources (digital or physical) and information.

◆ **Local area network:** a system that connects computers and other devices within a small geographical area, such as an office or home.

Welcome to **computer networks**. In recent decades, networks have become an all-pervasive and integral part of our modern lives. We use networks to:

- instantly communicate and collaborate with people around the world
- access a wealth of information, entertainment and services at our fingertips
- conduct business transactions, banking and online shopping with ease
- learn new skills, attend virtual classes and expand our knowledge
- remotely control and monitor our homes, cars and other connected devices
- share photos, videos and stories with family and friends in real time.

The power and ubiquity of computer networks have truly transformed the way we live, work and play. In this chapter, we will delve into the inner workings of these complex systems that have become so ubiquitous that we barely give them a moment's thought – except when things go wrong.

■ Local area networks (LAN)

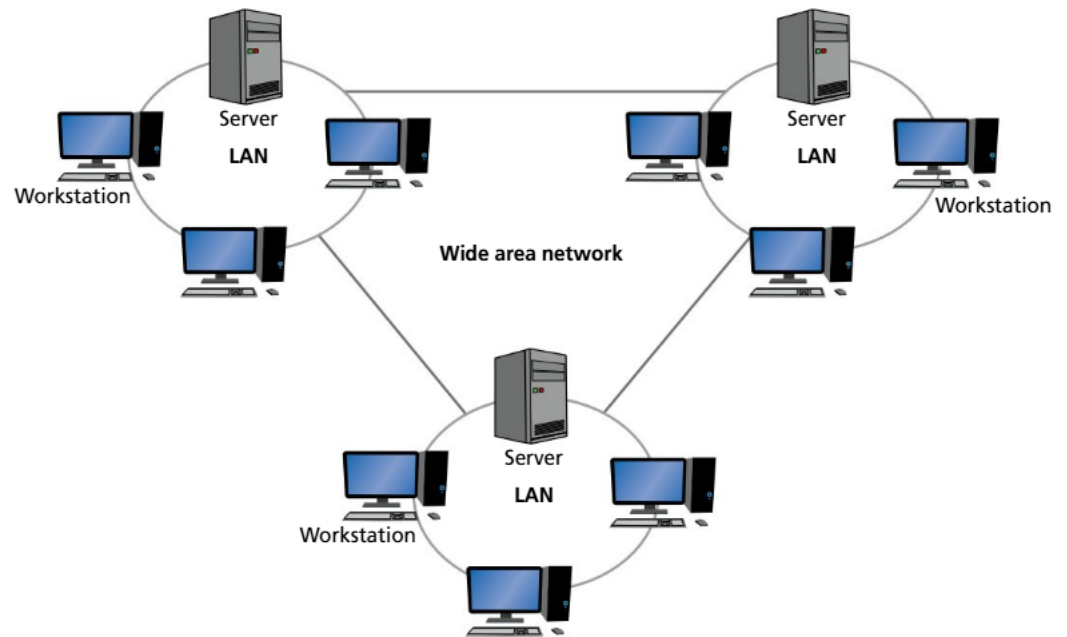
A **local area network** is a network of computers that are interconnected in a small geographical location, typically limited to a single property such as a home, building or campus. These are the oldest types of networks, though the equipment used in modern versions looks nothing like the historical versions.

The purpose of these networks is to facilitate sharing resources between the different computers, such as files, printers, applications and access to external networks, such as the internet.

LANs typically have a high bandwidth internally, with speeds ranging from 100 Mbps (Megabits per second) to 10 Gbps (Gigabits per second). Their small geographical range means there are typically no issues with latency (the time delay for data to transmit across the network).

Most homes and many corporate LANs now use a mix of wired Ethernet cables and wireless networking technologies. Dedicated wireless LANs may sometimes be referred to as WLANs.

■ Wide area networks (WAN)



◆ Wide area network:

a system that connects computers and other devices across a large geographic area, usually connecting multiple LANs together.

◆ Personal area network:

a network for personal devices within the range of an individual person, usually connected with Bluetooth.

A **wide area network** provides for the interconnection of multiple local area networks over a wider geographical distance. This connection distance may be across town, between different cities or even different continents. For example, this could be for sharing resources across different office branches of a company.

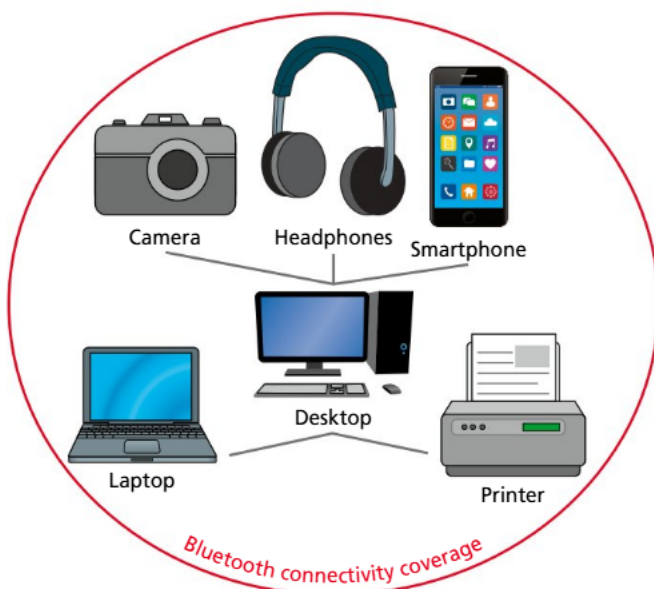
Classically, a WAN does not use the internet for this interconnection; instead it would have its own dedicated networking infrastructure for long-distance connections, such as fibre-optic cabling or point-to-point microwave transmission.

In practice, most modern WANs use the existing infrastructure of high-speed internet and establish a virtualized WAN through the use of virtual private networking technology, discussed below.

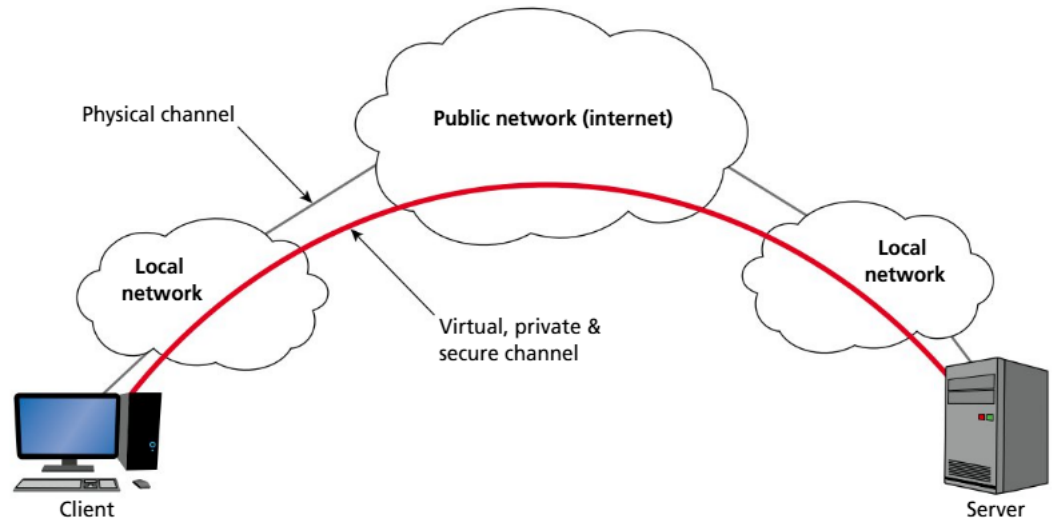
■ Personal area networks (PAN)

A **personal area network** refers to the devices that are interconnected, centred around an individual person. A PAN covers a very small range, and is typically limited to about 10 metres.

While USB cable-connected devices could be said to be part of a person's PAN, Bluetooth is the connectivity technology most commonly associated with PANs. The interconnected nature of your headphones, phone, camera, watch and whatever other devices you may be carrying on your person is what forms your PAN.



■ Virtual private networks (VPN)



◆ **Virtual private network:** a secure connection that runs across the internet to provide private communication between your network and a remote server.

◆ **Internet:** a global network of computer networks that are interconnected with each other and communicate through standardized protocols.

Virtual private networking refers to using public networking infrastructure to establish a secure, private tunnel for your own communication purposes. The encryption enables users to send and receive data across the public internet, as if their computing devices were directly connected on their own private network.

There are many “VPN companies” that advertise their services to retail consumers to make use of technology so that their internet browsing activities can appear to be taking place from a different geographical location from where they truly are. While this is a useful technology for getting around geo-blocks and the like, it is a different use case from the corporate use of VPNs.

Companies use the secure tunnel of a VPN to provide employees with remote access to their corporate networks as if they were physically present at their corporate headquarters.

VPNs help reduce the need for expensive dedicated networking infrastructure over long distances, as would historically have been required for a WAN.

A2.1.2 The purpose, benefits and limitations of modern digital infrastructures

■ Internet

To suggest the **internet** is a core component of modern digital infrastructure is a bit like saying water is wet. The internet is *the* global network. It connects millions of private, public, academic, business and government networks to one massive global network. It provides easy access to a vast amount of information and services, as well as facilitating near-instant communications. Commerce, businesses and global marketplaces are now extremely dependent upon it.

That said, the internet is far from perfect. In the 1970s and 80s, as key components of what would become the internet were being developed, there was no master plan, and the sheer scale of the result would have been impossible to imagine or foresee. This means there are vulnerabilities baked into the core technologies on which the internet depends. It is susceptible to hacking, denial of service, phishing and many other threats that we will look at later in this chapter.



Cloud computing

The cloud is a mysterious-sounding name for a critical part of modern infrastructure devised by a marketing major somewhere. In essence, it is computing services being made available for rent by large technology companies so that you don't need to purchase your own physical computer systems for servers and other infrastructure. Businesses and individuals can use the internet to access the software and hardware provided by these technology companies in their giant data centres.

There are some benefits to this approach. It means resource utilization can be easily scaled up or down without large financial investment. It also reduces the IT maintenance costs for small companies, as it is all part of the rented service.

Due to communication with cloud-rented systems occurring over the internet, that clearly means reliable, stable, high-speed internet connectivity is a must. Concerns about the security and privacy of data held on these third-party systems can also be a legitimate concern, as these systems are beyond your control.

Amazon Web Services, Google Cloud and Microsoft Azure are three big cloud providers, at the time of writing.



TOK

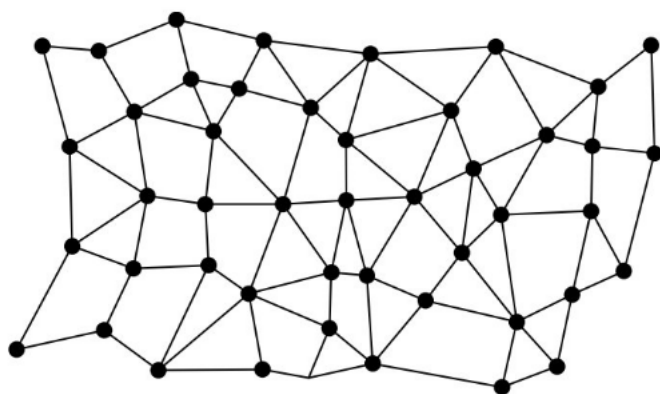


Role of experts in consumption or acquisition of knowledge

Computer networks, such as the internet, rely upon the expertise of network engineers to keep the system stable, efficient and effective. How has their contribution shaped the way we consume or acquire knowledge?

Linking question

How do cloud computing and distributed systems utilize networking to deliver services?



■ With each dot representing a node on the network, a distributed system has many different paths to traverse between any two points

Distributed systems

A distributed system connects multiple computers, or networks of computers, together to achieve common goals. They are more fault tolerant than other systems as, if one node fails, there are other paths of connectivity through which the rest of the network can continue to function. This resilience comes at the cost of complexity, meaning they are more difficult to design, manage and maintain. In particular, ensuring data consistency across the various nodes can be a challenge.

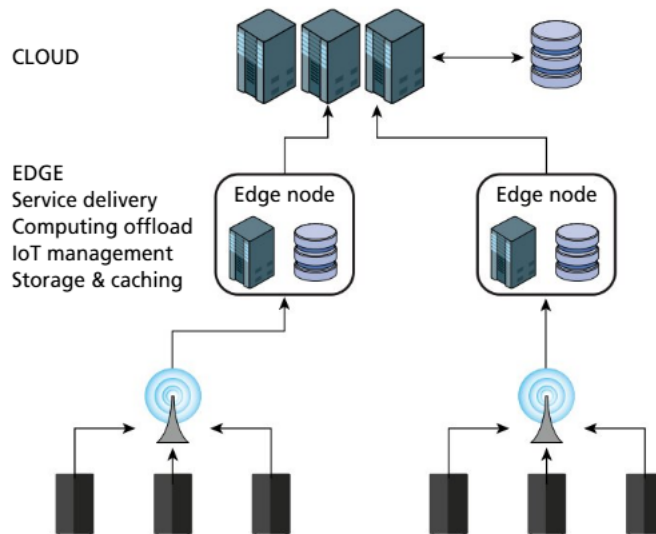
Peer-to-peer networking tools, such as BitTorrent, are examples of distributed systems, as are Blockchains. Distributed systems used by some large corporate networks include Content Delivery Networks to help distribute content to users worldwide.

■ Edge computing

Edge computing brings the computational and data storage capacity closer to the physical location where it is needed, to reduce latency and save bandwidth.

Large content delivery companies, such as Netflix, Spotify and the like, may use the edge computing model to deploy clusters of servers in key cities so as to reduce the amount of traffic that must travel back and forth to their main hubs. These servers can cache the most commonly sought-after files for each region, significantly speeding up the service for those customers, while preserving bandwidth for obtaining occasionally requested files that are not in the local cache.

This increase in endpoints to the network presents an increase in attack vectors for an adversary, so it comes with increased security and maintenance complexities.



■ Edge computing places computational resources closer to the point of demand, which reduces network load for the core servers in the cloud



■ Mobile networks

Mobile telecommunications networks are now globally ubiquitous. There are many parts of the world where mobile connectivity is the only connectivity available, as mobile networks can be deployed without the cost of digging up the ground for cabling installations.

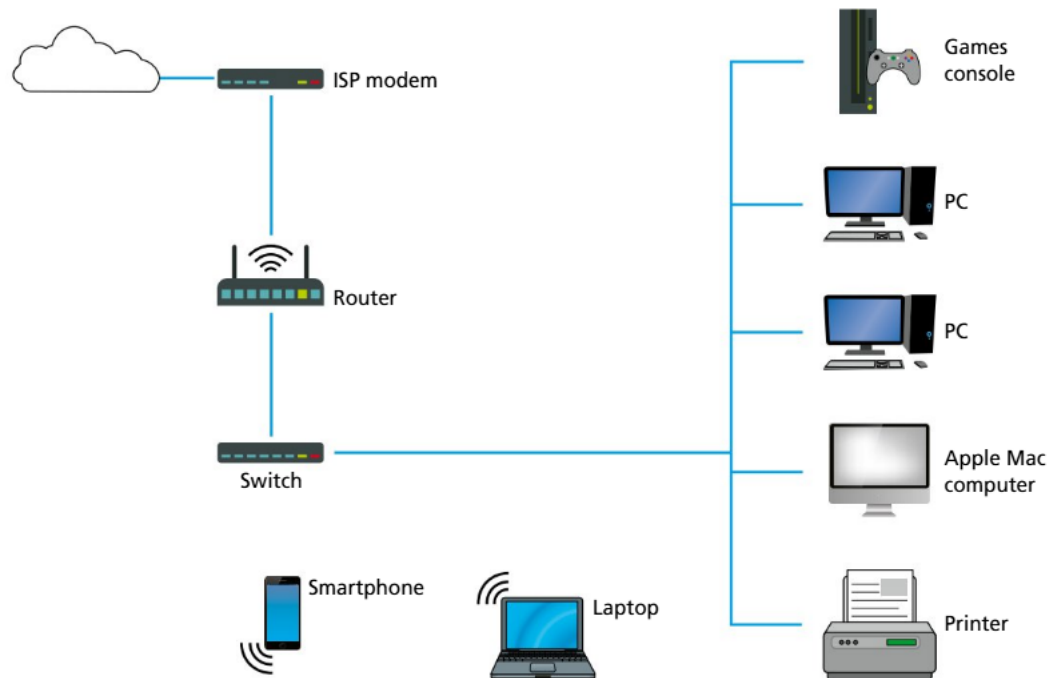
The mobility that these networks facilitate allows convenient access to information and communication while on the go, and supports vast regions. That said, the necessity to place towers everywhere and the complexities of geography do mean that mobile-phone networks do have *dead zones*, which are areas without network coverage. Signal strength can vary, which affects quality of service and bandwidth throughput.

At the time of writing, many countries are in the process of deploying fifth generation mobile networks (5G), which will increase speeds to up to 10 Gbps for peak data, with an average of 100 Mbps.



■ Mobile telecommunications tower

A2.1.3 The function of network devices



■ Devices in a typical home or small office network

Many home or small office networks might have only a single physical device that connects all their devices to their internet service provider. This single device might be colloquially referred to as the router or modem. The reality is that there are several logical devices at work, even if they are all co-located in a single physical unit. We will examine the roles of these different devices now.

● Top tip!

Understand the purpose!

There are a lot of different devices that serve different roles within a computer network. It can be easy to get them confused because most home networks have one physical device that plays many of these roles. Focus on understanding each device type's purpose and functionality. Ask yourself why this device type is needed, and what problems it solves.

■ Gateway

◆ **Gateway:** a device that connects different networks together and manages the traffic flow between them; often used to connect a local network to the internet.

A **gateway** is, as the name implies, the gate or connection between two networks. These networks may be communicating with two different protocols, and the gateway performs the translation task required to convert between the protocols.

For instance, in a typical home environment where the gateway connects to the internet service provider via a fibre-optic cable, it may be translating between EPON (Ethernet passive optical network) and regular Ethernet.

Because it converts from one network to another, it primarily operates at the application layer of the TCP/IP model (see Section A2.1.5).

◆ **Firewall:** a security system (hardware or software) that monitors and controls incoming and outgoing network traffic based on a set of security rules.

◆ **Router:** a device that forwards data packets between computer networks, routing the traffic along the most efficient path.

◆ **Network switch:** a device that connects multiple other devices within a single segment of a computer network, only forwarding data to the specific device it is intended for.

■ Hardware firewall

The hardware **firewall** monitors and allows or denies incoming and outgoing network traffic based on a predetermined set of security rules. The purpose is to be a safeguarding barrier between the locally trusted network and the untrusted network of the internet.

Firewall rules are typically a list of IP addresses and / or TCP/UDP ports to allow or deny traffic, based on the origin address or port and destination address or port.

The device functions at the transport and internet layers of the TCP/IP model (see Section A2.1.5).

■ Modem

A modem is a modulator–demodulator. It is used to convert between digital and analogue signals. A digital signal is modulated to encode digital data into an analogue form, transmitted across the analogue medium, such as phone lines, and then demodulated at the other end to extract the digital data.

The device functions at the physical, network interface layer of the TCP/IP model (see Section A2.1.5).

■ Network interface card

The network interface card is the hardware component on an individual device, such as a laptop or mobile phone, that allows it to connect to the network. It may be a card that requires connecting a physical cable, such as Ethernet twisted pair or fibre optic, or it could have an antenna attached for connecting to a wireless network.

The device functions at the physical, network interface layer of the TCP/IP model (see Section A2.1.5).

■ Router

The **router** directs the path packets of data take between networks. It inspects the network address information in the packet header to determine the ultimate destination and uses that to send the packet on the optimal route.

The router typically operates at the network interface layer of the TCP/IP model, as it uses TCP/IP addresses to make its routing decisions (see Section A2.1.5).

ACTIVITY

Thinking skills: A network router on a rocket flight has been accidentally reset, so that it is blocking legitimate traffic between the rocket and mission control. The task to fix it falls to you. Search Coding Quest and select “Broken firewall”, or go to <https://codingquest.io/problem/29>, to complete this activity.



■ Switch

A **network switch** connects devices within a single segment of a network. The switch effectively creates a network; it is the central spoke in a star-based network. A switch receives incoming packets of data and sends them on to their destination within the local area network using MAC addresses.

The switch operates at the network interface layer of the TCP/IP model (see Section A2.1.5).

■ Wireless access point

Wireless access points connect wireless devices together to form a network, in the same way that a switch does for a physical network. They can also act as range extenders for wireless signals.

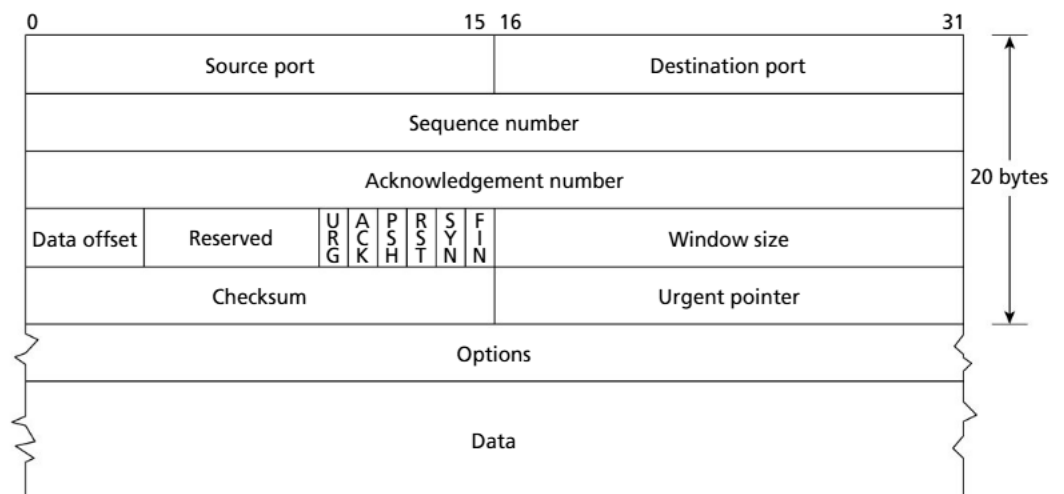
A wireless access point operates at the network interface layer of the TCP/IP model (see Section A2.1.5).

A2.1.4 Network protocols used for transport and application

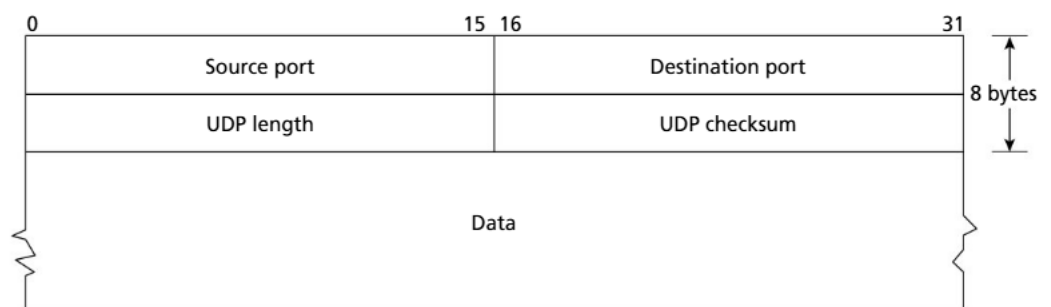
● Linking question

How do the concepts of binary and hexadecimal data structures relate to network communications? (B2)

■ Transmission Control Protocol (TCP) and User Datagram Protocol (UDP)



■ TCP protocol



■ UDP protocol

◆ **Protocol:** a set of rules and standards that define how data is transmitted and received across a network for a given application.

The Transmission Control Protocol (TCP) and User Datagram Protocol (UDP) operate on the transport layer of the TCP/IP model. That is, they are packets that are contained within the data portion of an IP packet. Where IP is responsible for getting a packet from one computer system to another, the TCP or UDP is responsible for getting data from one application to another.

These **protocols** facilitate multiple applications sharing a common network connection at once. For instance, if a server is running both an email-server application and a web-server application, the host operating system needs a means of determining which application to send inbound traffic to for processing. This is where port numbers come in. Think of these as phone extensions, or apartment numbers within a building address. To assist matters even further, the industry has standardized default port numbers for commonly used applications. The Simple Mail Transfer Protocol (SMTP) uses port 587 and Hypertext Transfer Protocol (HTTP) uses port 80, whereas Hypertext Transfer Protocol Secure (HTTPS) uses 443 and the Secure Shell (SSH) uses port 22.

TCP is a connection-oriented protocol. This means it establishes and maintains an active connection with the remote server until the application programs on both ends have finished exchanging messages. TCP ensures that data is delivered in order, without errors, and that data is acknowledged upon receipt. If there is an error in the receipt of a packet (either it doesn't arrive or is corrupted in some way), it is re-sent. The sequence number is used to ensure packets are reassembled into their correct order by the receiving application, and the checksum is used to ensure it is not corrupted during transmission.

UDP is a connectionless protocol. Data is transmitted, but there is no guarantee of reliability or order of delivery. UDP is used for applications such as streaming video, where maintaining speed and staying up to date with the broadcast is more important than the occasional missed frame or subset of pixels.

Sometimes a light-hearted approach can help communicate the difference between two items. Here is an anecdote that does the rounds of the internet comparing TCP and UDP:

A TCP joke:

Hello, would you like to hear a TCP joke?

Yes, I'd like to hear a TCP joke.

OK, I'll tell you a TCP joke.

OK, I'll hear a TCP joke.

Are you ready to hear a TCP joke?

Yes, I am ready to hear a TCP joke.

OK, I'm about to send the TCP joke. It will last 10 seconds, it has two characters, it does not have a setting, it ends with a punchline.

OK, I'm ready to hear the TCP joke that will last 10 seconds, has two characters, does not have a setting and will end with a punchline.

I'm sorry, your connection has timed out ... Hello, would you like to hear a TCP joke?

A UDP joke:

I know a UDP joke, but you might not get it.

For a server that receives requests, processes them and then replies with a response:

Python

```
import socket
def run_server(host="0.0.0.0", port=65432):
    # Create a socket object using IPv4 (AF_INET) and TCP protocol (SOCK_STREAM)
    with socket.socket(socket.AF_INET, socket.SOCK_STREAM) as s:
        # Bind the socket to the address and port, and start listening
        s.bind((host, port))
        s.listen()
        print(f"Server is running and listening at {host}:{port}")
        # Wait for connection. The code will pause here until connection
        conn, addr = s.accept()
        with conn:
            print(f"Connected by {addr}")
            while True:
                data = conn.recv(1024) # Receive up to 1024 bytes
                if not data:
                    break
                received_message = data.decode() # Decode bytes to string
                print(f"Received message: {received_message}")
                new_message = received_message.upper()
                print(f"Converted message: {new_message}")
                conn.sendall(new_message.encode()) # Encode to send as bytes
if __name__ == "__main__":
    run_server()
```

The following code is for a client that generates a request, sends it and then receives the response. You must ensure the server code (above) is running, and the code below is updated with the IP address of the server. You may need to adjust your firewall settings to allow this demonstration to work.

Python

```
import socket
def run_client(server_ip_addr, server_port=65432):
    # Create IPv4 and TCP socket
    with socket.socket(socket.AF_INET, socket.SOCK_STREAM) as s:
        s.connect((server_ip_addr, server_port))
        message = "Hello, server!"
        s.sendall(message.encode())
        data = s.recv(1024).decode() # Get response and decode from bytes to string
        print(f"Received {data} from the server")
if __name__ == "__main__":
    server_ip_addr = "127.0.0.1"
    run_client(server_ip_addr)
```

■ Hypertext Transfer Protocol (HTTP)

HTTP is the foundation of the world wide web. It is the protocol used for the transmission of hypermedia documents, such as HTML, as well as associated text files (Javascript, CSS) or binary files (images and so on).

HTTP is a stateless protocol. This means that each command is executed independently of any previous commands, with no recollection or knowledge of them. The client (such as a web browser) sends requests to the web server, which then responds with the requested resource or an error code.

HTTP communication occurs as unicode text strings, so it is human-readable.

An example of an HTTP request from a browser might resemble the following text. The client is sending a GET request for a particular file called `index.html` from the server hosting `www.example.com`.

```
GET /index.html HTTP/1.1
Host: www.example.com
```

The associated reply from the web server follows. The server begins by indicating it is replying using the HTTP `v1.1` protocol and a status code of `200`, which means OK. If there is an error then a different status code would be received. For instance, if the requested file was not found then a status of `404` would be sent.

```
HTTP/1.1 200 OK
Content-Type: text/html
Content-Length: 1456
<!DOCTYPE html>
<html>
<head>
  <title>Example Website</title>
</head>
<body>
  <h1>Welcome to the Example Website!</h1>
  <p>This is the content of the index.html web page.</p>
</body>
</html>
```

This is a simple example; in practice, there is also commonly authentication information, cookies and other data sent as part of the request or response.

You can programmatically experiment with sending your own HTTP requests and receiving the responses with Python by using the `requests` library. For instance, a simple program may look like this:

Python

```
import requests
response = requests.get("https://www.example.com")
print(response.status_code)
print(response.text)
```

The `requests` library is very powerful and even lets you download or upload text files and binary files.

If you'd like to create your own web server that receives and processes HTTP requests, take a look at Python's `Flask` library. A simple web-server application may resemble the following code:

Python

```
from flask import Flask, render_template, request, redirect, send_file
app = Flask(__name__)
app.config["SECRET_KEY"] = "code used to secure cookies from tampering"
# Return templates/index.html
@app.route("/")
def index_page():
    return render_template("index.html")
# Return a binary file
@app.route("/promotional_video")
def promotional_video():
    return send_file("promotional_video.mp4")
# Use the URL path to supply a parameter
@app.route("/user/<userid>")
def users(userid):
    return "User page for "+userid
# Get values from an HTML form
@app.route("/page2", methods=["GET", "POST"])
def page2():
    # HTML with <input name='person'> will create a request.values['person']
    form = dict(request.values) # Convert all values into a dictionary
    person = form["person"]
    return f"Hello, {person}, welcome to my website"
# Start the web server. These should be the last lines
if __name__ == "__main__":
    app.run(host="0.0.0.0", port=80, debug=True)
```

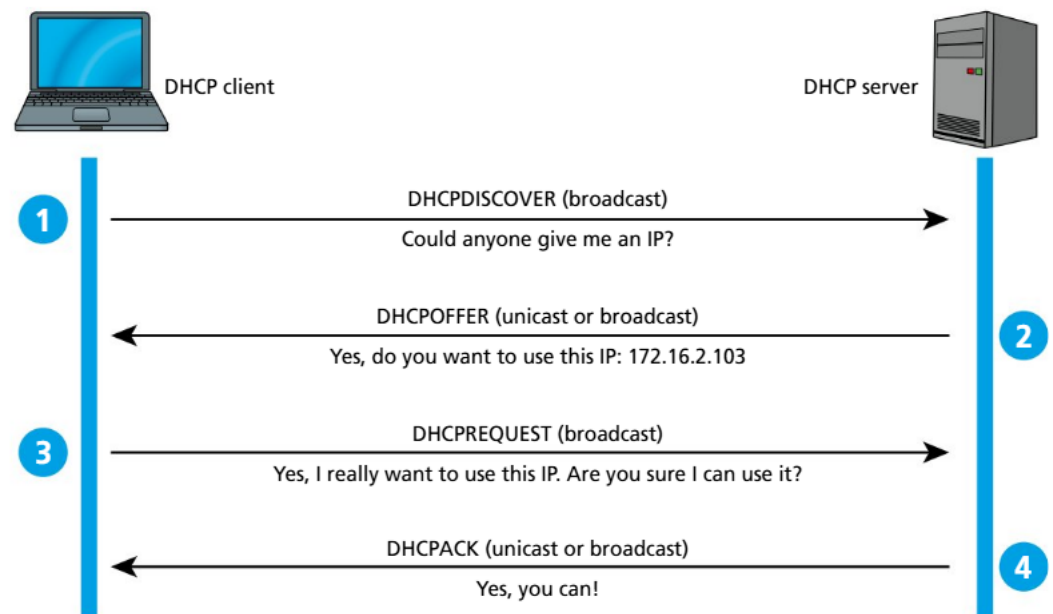
■ Hypertext Transfer Protocol Secure (HTTPS)

HTTPS is the secure version of Hypertext Transfer Protocol. It is designed to ensure communication is encrypted between the client and server to help protect against man-in-the-middle attacks and eavesdroppers. There are three key features that HTTPS offers over the original HTTP:

- **Encryption:** typically achieved through the use of either SSL or TLS encryption methods.
- **Authentication:** a digitally signed certificate is issued by the server so your client application can verify it is connecting to the correct server (i.e. to protect against an imposter posing as your bank website, for instance).
- **Data integrity:** ensures that the data sent and received is not altered in transit.

These ideas are discussed further in Section A2.4.4.

■ Dynamic Host Configuration Protocol (DHCP)



DHCP is a network-management protocol. It is used most commonly when a computer first joins a computer network (for example on system start-up). DHCP is the process by which the computer requests and receives a valid IP address for use on that network. In many home and small office networks, the device commonly known as the “router” also runs a DHCP server on it, providing IP addresses to devices that connect to it within the home.

The client device broadcasts (that is, sends a message for all devices on the network to see) a **DHCP Discover** message in which it provides its hardware MAC address and any preferences regarding subnet, router, DNS server or IP address lease time.

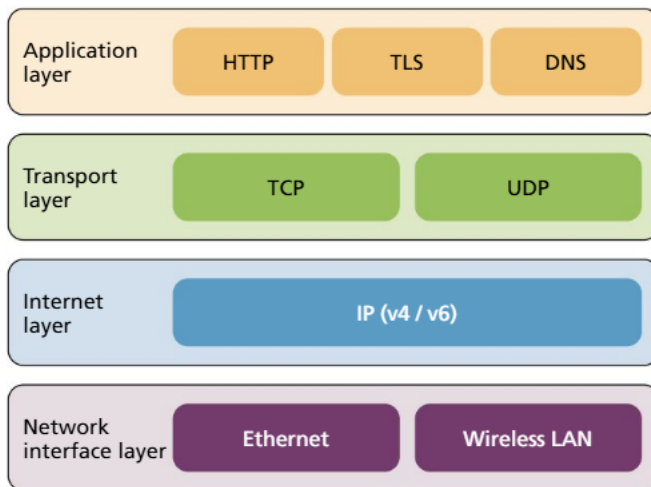
A DHCP server responds with a **DHCP Offer** message to assign an IP address from within a defined pool of addresses available to it. IP addresses are “leased” to devices for a specific period, after which the device must request to renew its lease. This process allows for reconfiguration and reuse of addresses. The offer also includes information about subnet, domain name server and other matters.

In the home or small office space, there are a number of addresses that have been reserved for private internal networks to use. These addresses are not used by valid servers and are not routable on the public internet. These address ranges are:

- 10.0.0.0 to 10.255.255.255
- 172.16.0.0 to 172.31.255.255
- 192.168.0.0 to 192.168.255.255

Networks that use these addresses for their internal devices rely on services such as network address translation to connect to the public internet. See Section A2.3.1 for a discussion about this.

A2.1.5 The function of the TCP/IP model (HL)



■ The allocation of bits in the TCP and UDP protocols.

The TCP/IP model is a conceptual approach to understanding the different roles and responsibilities of networking communication. It largely supersedes and streamlines the previously used OSI model.

The TCP/IP model is broken down into the four layers: the network interface layer (also known as the “physical layer”), the internet layer, the transport layer and, finally, the application layer.

■ Application

The application layer is the topmost level and is where the protocols used by actual applications reside. They include applications such as HTTP and HTTPS for web browsing; FTP for file transfer; SMTP and POP3 for email; and DNS. The application is responsible for using and correctly forming messages transmitted through these protocols.

■ Transport

On initiation of a communication request from an application, the operating system uses the transport layer to help ensure the data gets from the correct source application to the correct target application at the destination.

The two primary protocols that reside in the transport layer are TCP (Transport Control Protocol) and UDP (User Datagram Protocol).

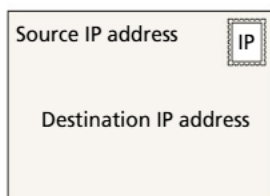
The transport layer receives data from the application layer, segments it and handles error detection and correction (if using TCP), along with retransmitting lost packets.

These segments or datagrams are then transferred to the internet layer.

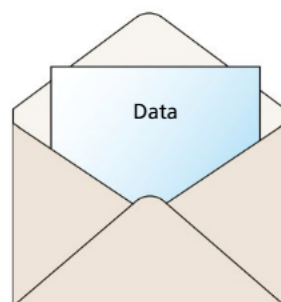
■ Internet

The internet layer is responsible for managing the movement of packets across the network, including ensuring they are routed to the correct destination.

To do this, it receives segmented data from the transport layer, encapsulates it into packets and decides on the best route for the packets to travel across networks. These packets are then forwarded to the network interface layer, which takes care of transmitting the bits to the next physical device in the chain.



■ TCP/IP data as packets



■ Network interface

The network interface layer manages the process of physical transmission of the data across the networking hardware and transmission media. It includes the protocols and hardware necessary to deliver the data across the local media, and aspects such as Ethernet, WiFi and physical network components including routers, cabling and switches.

The network interface layer receives data packets from the internet layer and converts them into a form suitable for transmission over the network, such as over Ethernet cabling or WiFi. Hardware addressing, such as MAC addresses, is managed by this layer.



Common mistake

It's common to confuse which protocols operate at which layer of the TCP/IP model. Keep a clear map of where TCP, UDP, HTTP, HTTPS and DHCP fit within the layers.

REVIEW QUESTIONS

- 1 Which network is typically used to cover a geographical area that spans a city or a group of buildings?
 - a PAN
 - b LAN
 - c WAN
 - d VPN
- 2 Which type of modern digital infrastructure would be most suitable for handling real-time data processing close to data sources?
 - a Internet
 - b Cloud computing
 - c Distributed systems
 - d Edge computing
- 3 Which device is responsible for enabling multiple computers to connect to the internet through a shared connection?
 - a Modem
 - b Switch
 - c Router
 - d Network interface card (NIC)
- 4 Which protocol ensures delivery of packets and is used for activities requiring high reliability, such as file transfers?
 - a TCP
 - b UDP
 - c HTTP
 - d DHCP
- 5 In the TCP/IP model, which layer is primarily responsible for routing and forwarding packets?
 - a Application
 - b Transport
 - c Internet
 - d Network interface
- 6 Describe one characteristic that distinguishes a virtual private network (VPN) from other types of networks.
- 7 Identify one major limitation of using cloud computing in digital infrastructure.
- 8 Describe the primary function of a wireless access point in a network.
- 9 Describe three main differences between HTTP and HTTPS.
- 10 Describe the role of the application layer in the TCP/IP model.

SYLLABUS CONTENT

By the end of this chapter, you should be able to:

- ▶ A2.2.1 Describe the function and practical applications of network topologies
- ▶ A2.2.2 Describe the function of servers (HL)
- ▶ A2.2.3 Compare and contrast networking models
- ▶ A2.2.4 Explain the concepts and applications of network segmentation

TOK

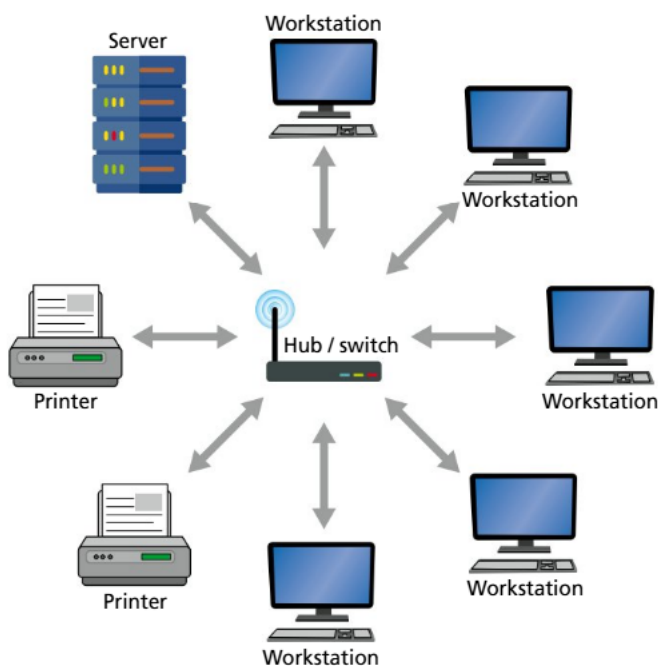


Organizing and classifying knowledge

If the sum of human knowledge is being stored on servers accessible through the global interconnected computer network known as the “internet”, how does the organization and classification of the internet, and its constituent networks, affect what we know?

Linking questions

- 1 How can network types, or transmissions, impact database performance? (A3)
- 2 How do network topologies influence machine learning algorithms (A4)?



■ Star topology has all the individual nodes emanating from a central point

A2.2.1 The function and practical applications of network topologies

Network topology refers to the physical layout and structure of the nodes and connections within a network.

■ Star topology

In a star topology, all nodes connect to a single central device (such as a network switch). The central switch manages the task of routing messages between the various nodes. This approach is very common in homes and small offices, where a single switch provides all the capacity needed for the various devices that are connecting.

A few factors to consider with the star topology include:

- **Reliability:** If the central device fails, the network fails.
- **Bandwidth speed:** While each device has its own dedicated connection to the hub, the overall throughput speed is dependent on the processing capacity of that central hub.
- **Scalability:** The central switch typically has a limited number of ports or connections that it can handle, possibly limiting expansion to accommodate future needs.
- **Collisions:** There is minimal risk of data collision as every node has its own dedicated connection.
- **Cost:** Quite low if only a few nodes are needed; however, costs can increase if the capacity limits of the central node are exceeded and upgrades are required.

Top tip!

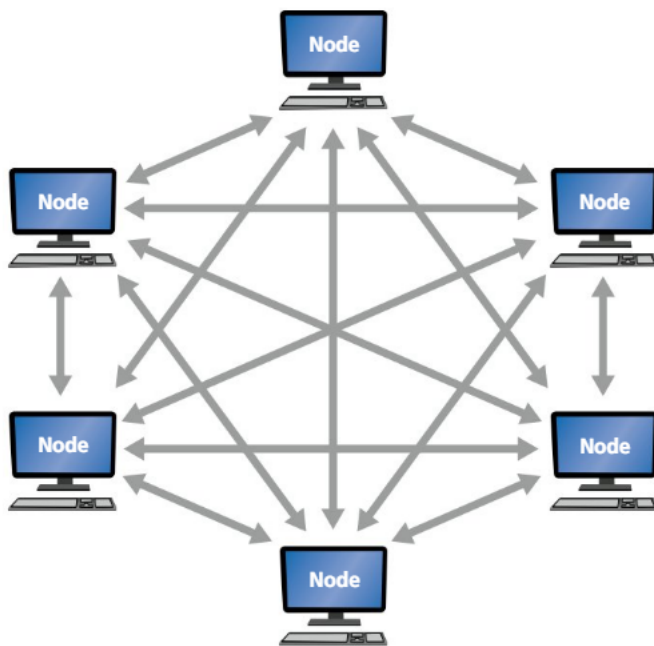
Each topology (star, mesh, hybrid) serves different needs. Be sure to understand why each one is more suitable than another for specific situations. Learn the diagrams of the different topologies to visualize how nodes are connected, as that provides a crucial understanding of how the flow of data is impacted if a node fails.

Mesh topology

In a mesh topology, every node has an immediate, direct connection to every other node.

While most often depicted as a full mesh, partial meshes do also exist.

Meshes are suitable for large environments where reliability of individual pathways may be a concern. Critical infrastructure, such as military and aviation environments, is more inclined to use a mesh topology to help ensure robust resilience of the network against any point of failure.



■ In a mesh topology, all nodes are interconnected to all other nodes

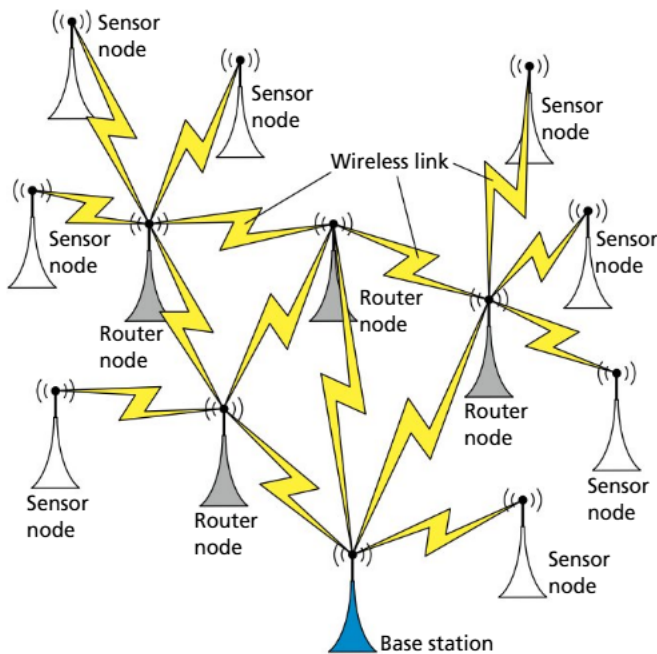
Wireless mesh systems, such as Meshtastic, have also started to gain popularity among tech hobbyists and as a means of providing connectivity in remote and disaster-prone areas.

Factors worth considering in a mesh environment include:

- **Reliability:** Highly reliable and robust against the failure of any individual node or communication path.
- **Bandwidth speed:** High, given that each node can communicate directly with its intended target node.
- **Scalability:** Adding new nodes is expensive as cabling and infrastructure must be run to all other nodes on the network.
- **Collisions:** Minimal risk of collisions due to the direct connections available.
- **Cost:** High due to all the additional cabling and networking infrastructure required for all the redundant connections.

Hybrid topology

The hybrid topology combines a mix of two or more other topologies. It makes maximum use of the advantages of each approach while minimizing the downsides. Hybrid approaches are suitable for large enterprises and telecommunications networks. In the illustration, a mesh



■ The hybrid topology is a mix of the mesh and star topologies; key points in the network are interlinked like a mesh, and then each of those points is the central point for its own star of connected nodes

approach is used for the interconnection of different base stations to provide reliability and durability between base stations, and then each base station manages a star topology for the individual devices connected to it.

Factors to consider for a hybrid approach include:

- **Reliability:** Generally quite high; if a node fails, it only affects the clients immediately connected to it, but not other parts of the network.
- **Transmission speed:** This is dependent on the exact configuration and mix of topologies deployed; bottlenecks can be an issue if it is not carefully designed.
- **Scalability:** Typically highly scalable and adaptable to changing needs and circumstances; a small addition to a network could start as a spoke on a star, and then later be upgraded to a full node in a mesh if needed.
- **Collisions:** Again, this is dependent on the configuration and mix of topologies.
- **Cost:** Likely to be higher to get started but generally more efficient over the long run.

A2.2.2 The function of servers (HL)

Top tip!

It is very common for one physical computer acting as a server to have many server-based applications running on it simultaneously. The one computer could have a web server on port 80 and 443; an FTP server on 20 and 21; an SSH server running on port 22; and a DNS server on 53. In this section, “server” relates more to the separate server software applications rather than separate physical devices.

◆ **Server:** a computer or device on a network that manages and provides various network resources on behalf of other computers (clients) on the network.

◆ **Domain name:** a human-readable name assigned to a specific IP address on the internet, e.g. www.example.com.

A **server** is a computer that provides a service to other computers or devices on a network. Typically, these are dedicated machines housed in a specialized room, centrally located on the network with backup power supplies and other systems.

This section looks at some of the common application servers frequently seen on networks. We will look at each from a perspective of their function, scalability, reliability and security.

Domain name server (DNS)

The **domain name** server provides a translation service that converts human-readable domain names into the IP addresses that are required for routing purposes. For example, at the time of writing, hachettelearning.com translates to 78.136.36.226.

The client device sends the DNS request to its configured DNS server. This can either be specified manually in the operating system settings, or it can be provided via DHCP when the

device connects to the network. If the DNS server that receives the request does not know the answer to the query, it asks its DNS server, and so on up the chain until a server can provide a response. The DNS server then typically caches that response for a while in case it receives the same query.

On Windows computers, the `nslookup` terminal command allows you to execute DNS queries, and on macOS machines use the `host` command. In Python, there is a built-in `dns.resolver` module that you can use.

Python

```
import dnspython as dns
import dns.resolver
domain = "www.example.com"
try:
    resolver = dns.resolver.Resolver()
    answer = resolver.resolve(domain, "A")
    for record in answer:
        print(record.to_text())
except dns.resolver.NXDOMAIN:
    print(f"The domain {domain} does not exist.")
except dns.resolver.Timeout:
    print(f"The DNS lookup for {domain} timed out.")
except dns.resolver.NoAnswer:
    print(f"No answer found for {domain}.")
except Exception as e:
    print(f"An error occurred: {e}")
```

- **Scalability:** DNS uses a distributed database and caching. This system allows for the global domain name system to manage with ease the billions of requests made per day.
- **Reliability:** DNS is a mission-critical system for the internet. Any downtime, particularly from the upstream servers, can affect many thousands or millions of client devices. With that in mind, the system is built with a lot of redundancy in place, where each server typically has at least three alternatives to query.
- **Security:** Given the global importance of DNS, it is frequently the subject of malicious behaviour. The key domain name servers around the world require extremely robust security systems including firewalls, intrusion detection systems and other countermeasures.

■ Dynamic Host Configuration Protocol (DHCP)

As previously discussed in Section A2.1.4, DHCP is responsible for assigning IP addresses and network settings to devices that request them on the network.

■ File server

A file server provides a centralized location to store, access and manage files. There are several commonly used approaches for this.

FTP (file transfer protocol) or SFTP (secure FTP) are protocols that are commonly used for accessing remote systems for the transfer of files. FTP usually involves opening dedicated FTP transfer utility software to perform the task.

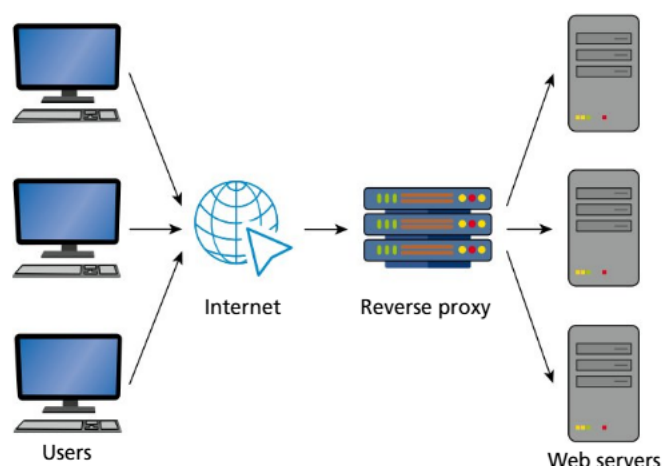
For an office environment, it is usually preferential for the storage of a file server to be present within the file, folder and disk structure of the local computer operating system as a mounted drive letter (Windows) or folder (macOS). To do this, protocols such as Server Message Block (SMB), Network File System (NFS) or Apple Filing Protocol (AFP) are more commonly used.

- **Scalability:** File servers are typically scaled through adding drives to provide additional storage, and upgrading the speed of the network interface cards when they are shared between many client devices.
- **Reliability:** RAID (redundant array of independent disks) is a technology that allows for combining multiple physical disk drives into one logical unit. This can be used to pool the storage together to create one larger drive (for example two 1 TB drives pooled together to present as if they were a 2 TB drive), or to provide backup redundancy in the case of one drive failing. RAID also allows for a hybrid approach to mix both functions together.
- **Security:** Access permissions need to be carefully managed to protect against unauthorized access or alteration of data. These access permissions can typically be set at a file or folder level, and can have either read-only permission or read-write permission.

■ Mail server

The mail server stores emails for local users on a network, and exchanges them with other mail servers when users send an email. Commonly used protocols include the SMTP (Simple Mail Transfer Protocol) and POP3 (Post Office Protocol).

- **Scalability:** Mail servers need to handle large volumes of emails and attachments efficiently. Many companies contract out their mail systems to third-party providers to help manage the demand.
- **Reliability:** As a critical system, email servers typically employ such mechanisms as queues and redundant systems to help maintain reliability.
- **Security:** Strong security systems for email are essential, given the frequency with which email is used as an attack vector for spam, phishing, pharming, malware attachments and other security threats. Best practice now involves the use of email authentication protocols such as SPF (Sender Policy Framework) and DomainKeys Identified Mail (DKIM). SPF allows domain name owners to specify which email servers are authorized to send emails on their behalf. DKIM allows domain name owners to sign their emails digitally with a cryptographic signature that can be checked by the recipient to ensure the email has not been tampered with en route.



■ Reverse proxy server

■ Proxy server

In the business and commercial world of contracts, a proxy is a person you authorize to act on your behalf.

In that vein, a proxy server is traditionally used to act as the client device on your behalf to browse the greater internet. Its caching functionality is useful as it means it can remember content for you from different addresses and return that request immediately without having to generate additional external traffic. Organizations such as schools and offices may use a proxy server for requests coming from their internal network to reduce costs. As the request being received by a server appears to have originated from

the proxy and not the actual client, students may also use proxy servers installed on computers at home as a means of getting around school network filters.

Proxy servers can also be configured to act on behalf of the server rather than just client devices. This arrangement is known as a “reverse proxy”. This arrangement is typically used to help distribute the load of incoming requests among a range of servers, and also to cache the common responses given to save calculation and processing load on the servers themselves.

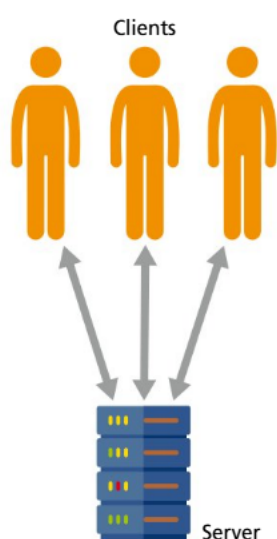
- **Scalability:** Reverse proxy configurations can dramatically improve the scalability of a web server to handle significantly more traffic than would otherwise be the case.
- **Reliability:** When used to balance loads, a proxy server provides fault tolerance capacity.
- **Security:** A proxy server provides an additional layer of security by shielding internal client devices from direct exposure to the internet.

■ Web server

A web server hosts web pages and related content, and serves them to client devices across the internet. The web pages can either be static documents (such as HTML) or dynamically produced at runtime from programming code, in which case the web server sends the request to the application in the style of a reverse proxy, and then sends the reply back to the original client.

Nginx is a popular open source web server.

- **Scalability:** Modern web servers are quite efficient but, when demand for their services is high, the load is typically distributed across multiple servers, with a reverse proxy server acting as the public interface.
- **Reliability:** If the web server fails, then all websites it is hosting also fail. As web-based applications become critical infrastructure for some organizations, the importance of reliable web server software grows.
- **Security:** A web server needs to be secured against attacks such as distributed denial of service. Additionally, modern web servers provide encryption and authentication through SSL to run HTTPS, rather than just HTTP traffic, for a safer browsing experience. Let's Encrypt (<https://letsencrypt.org>) provides a free service for obtaining the SSL certificates needed to offer HTTPS browsing through a web server.



A2.2.3 Networking models

■ Client-server

The client-server model is where devices take on the role of either being the client that requests a network service, or the server that provides a network service.

Benefits:

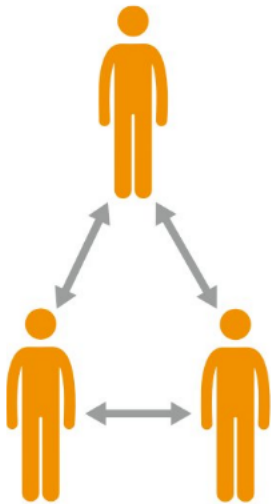
- **Centralized control:** It is easier to manage and update systems from a central point.
- **Scalability:** It is easier to add a more powerful server (for example with increased processing, memory or storage) if it is a central system.
- **Efficiency:** A server can be optimized for a specific task.
- **Security:** It is much easier to manage security risks if everything is run from a central point.

Drawbacks:

- **Single point of failure:** If the server goes down, all services attached to it fail.
- **Single point of risk:** If the security of the server is compromised, all data and services managed by the central point are also exposed and vulnerable.
- **Cost:** Creating a central point to manage all the requests means a significant investment in expensive hardware and software to cope with that demand.

Real-world applications:

- **Web browsing:** Clients (browsers) request web pages from servers.
- **Email:** Email servers manage the sending and receiving of emails to and from client applications.
- **Online banking:** Central servers handle transactions, authentication and data storage, offering high security and reliability.



■ Peer-to-peer

In the peer-to-peer model, there is no server that co-ordinates the network services; each device is both a client and a server, and communicates with all other peers. The provision of network services is distributed across the devices.

The challenge with peer-to-peer is co-ordination, since there is no obvious “authoritative” source to query where to find certain services.

Benefits:

- **Decentralization:** There is no single point of failure or risk.
- **Cost effective:** There is no need to spend money on expensive hardware or software, as the load is shared among all the clients themselves.
- **Scalability:** As the size of the network grows, so too does the scale of services it can provide. Each additional peer also adds additional capacity.
- **Direct sharing:** Services can be provided directly from one computer to another, reducing bottlenecks.

Drawbacks:

- **Co-ordination:** Maintaining settings, security patches and synchronization of data across multiple nodes on the network can be very challenging.
- **Reliability:** Since services are being provided by peers, which are just client devices on the network, if someone turns their device off, then it is no longer providing whatever services it was supplying to the network. Availability of systems and resources on the network can vary without much notice.

● Common mistake

Networking models

Don't oversimplify the distinction between client-server and peer-to-peer models. Understand the nuances involved that might make one more suitable than the other for specific scenarios. Failing to consider scalability and maintenance aspects can lead to incomplete understandings.

Real-world applications:

- **BitTorrent:** Files are shared directly between users without a co-ordinated central hub.
- **Voice over IP:** The data for calls and video conferencing are routed directly from client to client, rather than creating massive data bottlenecks for a centralized server.
- **Blockchain:** Bitcoin and other blockchains operate decentralized databases across many nodes to help guard against tampering with the transaction ledger.

A2.2.4 Concepts and applications of network segmentation

◆ **Network segmentation:** dividing a computer network into smaller, distinct subnetworks to improve performance, security and management.

Network segmentation refers to logically splitting a larger network into smaller, more manageable parts. Each of these segments can be isolated from each other to provide better security, performance or management of system resources.

Two commonly used approaches for network segmentation are subnetting and virtual local area networks (VLANs).

Subnetting divides a network into subnetworks based on unique ranges of IP addresses. For instance, a school may place staff devices on 192.168.0.X, and student devices on 192.168.1.X. Each subnet can share the same networking infrastructure, such as wireless access points, switches and cabling, but would not “see” the devices on the alternate subnet. This approach allows devices to be logically grouped based on department, function or geography, and can allow for different security policies to be applied, depending on the organization’s needs.

Virtual LANs are another way of separating one physical network into two logical networks without requiring duplication of infrastructure. They create distinct broadcast domains that are mutually isolated unless explicitly allowed to communicate via routing. Traffic is isolated to members of the VLAN only. Where a client device may change its subnet by manually changing its IP address, VLANs are more secure and typically use MAC addresses or login credentials to determine which VLAN a device gets connected to. Because a virtual LAN is a logical construct rather than a physical one, it is worth noting that a VLAN could extend over numerous physical networks (such as over a VPN link).

Whichever method is used, segmentation reduces overall network traffic within each segment, allowing for more efficient data transmission. Network congestion for high bandwidth applications can be contained to an individual network segment, resulting in the other segments within the network being unaffected. Security is also significantly enhanced, as visibility of servers and their services can be restricted to particular VLANs.

Common mistake

Network segmentation

Mixing up segmentation, subnetting and VLANs is common. Make sure you define and distinguish each clearly.

REVIEW QUESTIONS

- 1 Which network topology is most beneficial in a large campus setting due to its scalability and robustness?
 - a Star
 - b Mesh
 - c Hybrid
 - d Ring
- 2 Which type of server is responsible for translating domain names into IP addresses?
 - a DHCP server
 - b DNS server
 - c File server
 - d Web server
- 3 Which networking model typically involves one or more central servers that manage data and resources for client devices?
 - a Client-server
 - b Peer-to-peer
 - c Hybrid
 - d None of the above
- 4 What is the primary purpose of implementing network segmentation within a corporate environment?
 - a To increase the number of devices on the network
 - b To enhance security and performance by reducing congestion
 - c To eliminate the need for routers
 - d To simplify network management
- 5 Outline three benefits of using a star topology in a home office setting.
- 6 Describe one function of a proxy server for each of:
 - a a large university campus
 - b a personal home.
- 7 Outline how a reverse proxy functions, in contrast to a normal proxy. What are its benefits?
- 8 Explain a key disadvantage of using a peer-to-peer network model for online banking.
- 9 Explain how VLANs contribute to network segmentation.

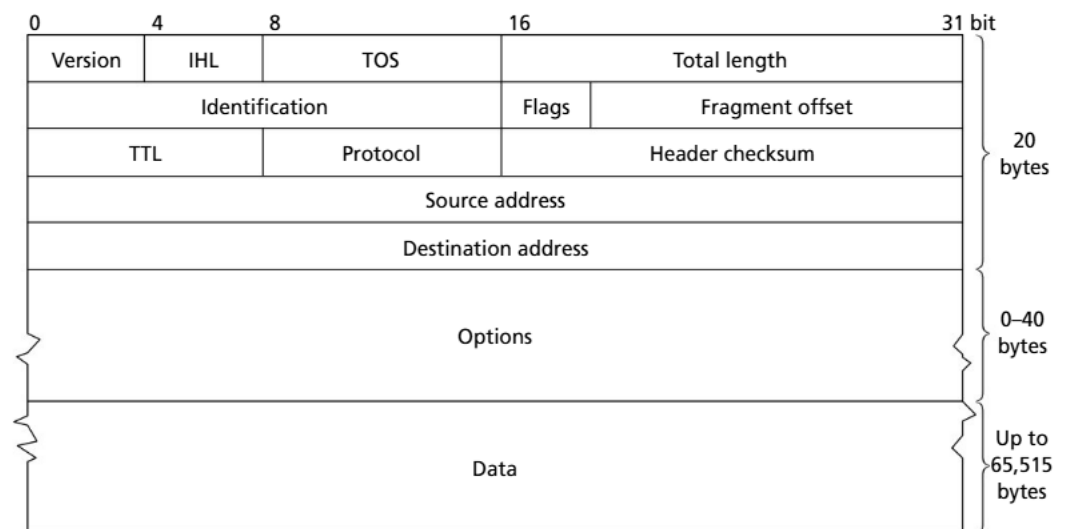
SYLLABUS CONTENT

By the end of this chapter, you should be able to:

- ▶ A2.3.1 Describe different types of IP addressing
- ▶ A2.3.2 Compare types of media for data transmission
- ▶ A2.3.3 Explain how packet switching is used to send data across a network
- ▶ A2.3.4 Explain how static routing and dynamic routing move data across local area networks (HL)

A2.3.1 Types of IP addressing

TCP/IP version 4 and version 6



■ Structure of IP version 4

Linking question

Are similar ethical principles needed when transmitting data over a network as when using data in machine learning algorithms? (TOK)

Common mistake

IP addressing

Don't mix up the address constructions of IPv4 and IPv6. Be sure to know the length and formatting of each version of IP addresses.

◆ **IP address:** a set of numbers that uniquely identifies each computer based on the Internet Protocol (either version 4 or version 6).

The key difference between IP version 4 and version 6 is the size of the address space. IPv6 was designed to deal with the long-anticipated problem of IPv4 addresses running out.

IPv4 uses a four-byte or 32-bit address, which is why it is typically displayed as four numbers separated by dots, where each number is in the range of a byte, 0–255, for example 192 . 168 . 0 . 1. Given there are 2^{32} possible addresses using these numbers, it means IPv4 is limited to approximately 4,300,000,000 nodes on the network.

IPv6, on the other hand, uses 16 bytes or 128 bits for its addresses. These addresses are typically displayed as eight groups of four bytes written in hexadecimal form, such as 2001:0db8:85a3:0000:0000:8a2e:0370:7334. Having 2^{128} addresses equates to a capacity for a staggering 340,282,370,000,000,000,000,000,000,000,000 nodes.

■ Public and private addresses

The key difference between public and private addresses is their visibility on the public internet.

IP addresses that are used on the public internet must be globally unique. They are typically assigned to websites, external-facing servers and routers that connect to the internet. The allocation of these public internet addresses is managed by ICANN (the Internet Corporation for Assigned Names and Numbers).

Private IP addresses used within a private network are not directly routable on the global internet. They are commonly used in home and corporate networks for devices such as computers, tablets and internal servers. Traffic from these devices must be converted to a public address by the router to get on to the public internet.

ICANN has reserved certain ranges of addresses as private, non-routable addresses for organizations to use for their internal networks. These are:

- 10.0.0.0 to 10.255.255.255
- 172.16.0.0 to 172.31.255.255
- 192.168.0.0 to 192.168.255.255
- IPv6 has FC00:0000:0000:0000:0000:0000:0000:0000 to FDF5:FFFF:FFFF:FFFF:FFFF:FFFF:FFFF:FFFF.

■ Static and dynamic addresses

Static IP addresses are those that are permanently assigned to a device. The device uses the same IP address every time it connects to the network. Static IP addresses are essential for servers so that client devices know where to locate them on a network.

In contrast, a dynamic IP address is one that is assigned to a device when it connects to the network. This process is overseen by a DHCP server that allocates connecting devices an address from a predetermined pool of addresses that it has available to it. Dynamic addresses help reduce overheads by more efficiently utilizing a limited number of IP addresses in environments where devices frequently connect and disconnect.

■ Network address translation (NAT)

◆ **Network address translation:** modifies the IP addresses of data packets as they pass through a router or firewall; this helps improve security and manages the limited number of IP addresses available through IPv4 by allowing multiple devices to share a single global IP address.

● Top tip!

Network address translation

Make sure you have an appreciation of, and recognize the role of, **network address translation** (NAT). Its capacity to allow multiple devices on a private network to share a single IP address for internet communications has kept the internet as a viable, functioning system. Its role in conserving global IP address space cannot be overstated.

Network address translation is a process whereby networks translate private IP addresses to public IP addresses, and vice versa. The process enables multiple devices on a network to share a single public IP address.

NAT has been crucial for the effective operation of the internet as it still remains highly dependent on IPv4 addresses, which have been all but exhausted. It also helps provide an additional layer of security for home and small office networks by hiding internal IP addresses from the external network.

A2.3.2 Types of media for data transmission

Top tip!

Side by side

Create a side-by-side comparison table for fibre-optic, twisted-pair and wireless media, with a focus on bandwidth, cost and installation complexity. Consider the roles for each type, such as fibre optic in data centres, twisted pair in offices and wireless for campuses, homes and mobile set-ups. Pay special attention to how security varies across the different media, particularly the susceptibility to eavesdropping or interference.

■ Fibre-optic cabling

Advantages:

- **Bandwidth:** Fibre optic has the highest data-transmission rates of the media under consideration in this section. Active research continues to make it even better, but current technology already allows for terabits per second to be transmitted.
- **Interference susceptibility:** As fibre-optic cabling uses light waves to transmit data, it is not susceptible to electromagnetic interference, making it ideal for environments with electrical noise, such as that generated by motors.
- **Range and attenuation:** Fibre-optic cabling is the medium of choice for the intercontinental submarine cables that run across the ocean floor, due to its very low signal loss over long distances. Commercially available single-mode fibre-optic cables can easily run distances of tens or hundreds of kilometres before a repeater is required.
- **Security:** Fibre-optic cables are very difficult to “tap” without detection, making them an excellent option where security is a concern.

Disadvantages:

- **Cost:** Fibre-optic cables are generally more expensive than other forms of cabling, both with respect to the cabling materials and supporting hardware infrastructure and the installation labour time.
- **Reliability:** Fibre-optic cables are more fragile than metallic cables and are prone to damage if improperly handled. Specifically, they have a limited radius for the maximum bend they can safely operate under.
- **Installation complexity:** The issues regarding supporting infrastructure requirements and proper handling of the cables themselves tend to result in fibre-optic cables requiring more specialized skills and equipment for installation and maintenance operations.

■ Twisted-pair cabling

Advantages:

- **Cost:** Twisted-pair cabling has been a well-known and established technology for decades now. It is widely and cheaply available.
- **Installation complexity:** Installation is considerably less complex than for fibre optic. A do-it-yourself approach is frequently taken for installation of simple home and small office networks, with telephony technicians typically only required for outdoor and other complex environments.
- **Reliability:** The cables are vastly more flexible, meaning they can be safely and reliably installed into tight spaces.

Disadvantages:

- **Bandwidth:** Twisted-pair cabling is typically an order of magnitude slower than the equivalent in fibre optic. That said, it is still quite adequate for most home and small office arrangements, offering Gigabit bandwidth performance. Twisted-pair cabling also significantly outperforms wireless networks. The highest quality twisted-pair cable currently available – Category 7 Ethernet – is rated for speeds of up to 10 Gb/s.
- **Interference susceptibility:** Given it is a cable that uses electrical signals to transmit data, twisted-pair cabling is quite susceptible to electromagnetic interference from the surrounding environment. This is particularly the case where shielding has not been wrapped around the cables.
- **Range and attenuation:** The range of twisted pair is considerably less than fibre optic. Category 6a twisted-pair Ethernet cables typically only have a range of approximately 100 metres.
- **Security:** As it is easier to tap into than fibre-optic cables, twisted pair poses a more significant security risk; however, an attacker must still get in close physical proximity to the cable, unlike with wireless.

■ Wireless transmission

Advantages:

- **Installation complexity:** There are no physical cables, so installation complexity and cost are significantly lower than for physical cable media.
- **Cost:** Wireless networks can be a cost-effective means of covering a large area where cabling is not practical.
- **Reliability:** Wireless networks provide mobility for devices to connect from, whatever the location, provided it is within the signal range. That said, reliability can be impacted through radio interference, discussed below.

Disadvantages:

- **Bandwidth:** In practical scenarios, the total available bandwidth is shared among connected devices, so the individual device throughput varies based on network conditions and the number of devices connected. Typically, a total bandwidth capacity of several Gb/s is available for the devices to share. This is lower than the equivalent twisted pair or fibre optic, and these also come with their own dedicated bandwidth, rather than a shared connection.
- **Interference susceptibility:** Wireless is very susceptible to interference from other wireless or radio signals, and physical obstructions can significantly affect signal quality.

Top tip!

Is your microwave safe?

If your home WiFi drops out when someone is using your kitchen microwave oven, it means it is leaking microwaves and should be urgently repaired or replaced! Microwave ovens operate at 2.4 GHz, which is one of the two frequencies used by WiFi (the other being 5 GHz). The reason microwaves use 2.4 GHz is that it is the resonant frequency of water so, by emitting high-energy waves, it excites the water molecules in your food, which causes it to heat up.

WiFi technology is considered safe to use, even though it also relies on 2.4 GHz, as the strength of the radio signals is measured in milliwatts, as opposed to most microwave ovens, which are pumping 800 to 1000 watts of energy into the surrounding environment.

- **Range and attenuation:** Range is typically limited to a maximum of about 90 metres in clear, line-of-sight conditions. Strength and quality of the signal deteriorate with distance and physical obstructions, such as walls.
- **Security:** Wireless technologies have a long and complicated history of being security risks since an attacker does not have to be physically present at a network connection to intercept the signal. Often, attackers can sit in a car outside the home or office and collect all your WiFi signals over the air. A lot of effort has been made to add layers of encryption to modern WiFi to mitigate this problem, but risks still exist.

A2.3.3 Packet switching

Top tip!

Account for packet loss

When considering the workload associated with **packet switching** and routing, be sure to remember that not all data transmitted is received at the other end. The seeming reliability of the modern internet is actually only due to all the error checking that occurs behind the scenes accounting for packet loss. This can have a significant impact on the performance of a network, especially where electromagnetic or other interference sources may abound.

◆ **Packet switching:** a method of sending data in small blocks, known as “packets”, across a network. Each packet can take a different path to reach its destination.

Common mistake

Confusing routers and switches

Ensure you don’t misunderstand the functional differences between routers and switches in network data transmission.

Packet switching is the foundation of most modern networking technologies, including the internet. It involves breaking large pieces of data into smaller, more manageable chunks called “packets”. These packets are transmitted across the network independently of each other and reassembled into the original whole at the destination. Packet switching is a highly scalable tool that allows efficient use of network resources, in contrast to older technologies such as circuit switching, where an entire dedicated channel or cable was used for the duration of a communication session (which deprived other devices of being able to use it).

The full process includes:

- **Segmentation:** Data (such as a file or email) that needs to be sent across the network is broken into smaller pieces known as “packets”. These packets contain a chunk of the data, plus information known as the “header” that contains source and destination network addresses and other control information. The MTU (maximum transmission unit) size of a packet is typically 1500 bytes, including header information.

- **Packet header:** The header of each packet contains information that helps ensure the efficient and reliable transmission of the packet across the network, such as the source and destination IP addresses. Additionally, each packet contains a sequence number that is used to reorder the packets at the destination into their correct sequence, so that the content of the data file is not jumbled. The header also contains error-checking information such as checksums, which calculate whether the packet has arrived correctly, or whether the source has to be asked to retransmit it.
- **Routing:** Each packet is sent through the computer network independently of any of the other packets associated with it. This means that packets for the same file may travel different paths across the network, based on conditions such as congestion and route availability. This flexibility in routing helps networks optimize usage and provide for a robust environment that is more tolerant of failures within the network.
- **Routers:** These devices determine the optimal path for each packet passing through to reach its destination. They inspect the packet destination address and use routing tables and algorithms to decide the next hop in the network to forward each packet to.
- **Switches:** These devices are typically operated within local area networks and direct packets between devices on the same network. They use the MAC addresses to process and forward packets.
- **Reassembly:** Once the packet has reached its destination, it is checked for sequencing information and then reassembled into the original file in the correct order. Importantly, because no two packets are guaranteed to be transmitted along the same path, it is impossible to assume that packets have arrived in the correct order. Additionally, some packets may have been lost or corrupted in transmission, so the source needs to be requested to retransmit those packets.

A2.3.4 Static routing and dynamic routing in local area networks (HL)

◆ **Routing:** the process of selecting paths along a computer network to send network traffic, based on the routing table, network performance and protocols.

Network **routing** refers to the algorithms that determine the path for traffic in a network, or between and across multiple networks. Within local area networks, routing decisions can be handled by one of two methods: static or dynamic routing.

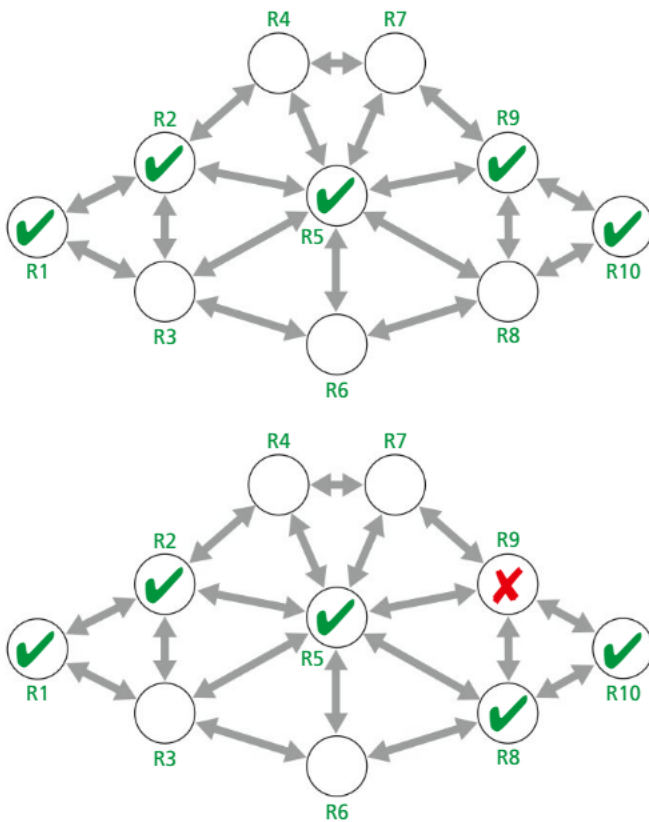
■ Static routing

Static routing is where the network routes are manually configured and entered into a routing table by the network administrator. These routes do not change except by manual update.

This method is easy to implement in small networks where the routes do not change much, but can be too complex to manage in networks with hundreds or thousands of nodes. Static routing provides predictability in the behaviour of your network, and does not require additional processing or bandwidth due to one router communicating with another for route discovery and optimization.

In addition to the challenge of scale, other issues around static routing include the lack of fault tolerance as, if a connection fails, static routing does not automatically attempt to find a workaround; and the maintenance workload associated with it, as every small change in the network requires manual updating of the routing table.

Dynamic routing



■ Dynamic routing coping with node failure

As can likely be inferred, dynamic routing contrasts with static routing by using an algorithmic approach to automatically adjust routes in the routing table. To do this, routers communicate with each other to share information about their connections, topology and demand on their network resources. This information is used to adjust the routes in runtime, with the goal of ensuring optimal paths for network efficiency.

Dynamic routing scales well for larger networks, as they can automatically adapt as changes occur. This also improves fault tolerance, as network traffic can be automatically re-routed around a failed connection.

Dynamic routing does pose some challenges. Its dynamic nature requires additional computational and bandwidth resources to manage. The combination of the larger networks and the dynamic routing do make for a more complex routing table that needs to be configured and maintained. This necessitates a deeper understanding of network concepts by the staff maintaining such a network. Finally, given each router is maintaining its own copy of a routing table, there can be delays in the convergence of the routing table while the individual routers communicate and negotiate with each other. This can lead to occasional routing inconsistencies in the short term.

REVIEW QUESTIONS

- 1 Which type of IP addressing is specifically designed to provide a larger address space than its predecessor?
 - a IPv4
 - b IPv6
 - c Static IP
 - d Dynamic IP
- 2 Which type of data-transmission media is most susceptible to electromagnetic interference?
 - a Fibre optic
 - b Twisted pair
 - c Wireless
- 3 What role do routers play in the process of packet switching?
 - a They prevent packet collision
 - b They direct packets along the most efficient paths to their destination
 - c They combine packets into a single data stream
 - d They generate the data packets from the user data

- 4 Which type of routing is better suited for networks that require frequent updates due to topology changes?
 - a Static routing
 - b Dynamic routing
 - c Both are equally suited
 - d Neither is suitable
- 5 Describe the main purpose of network address translation (NAT).
- 6 Describe one advantage and one disadvantage of using fibre-optic cables for data transmission.
- 7 Explain why packet switching is considered efficient for data transmission over a network.
- 8 Explain one main advantage of static routing compared to dynamic routing.

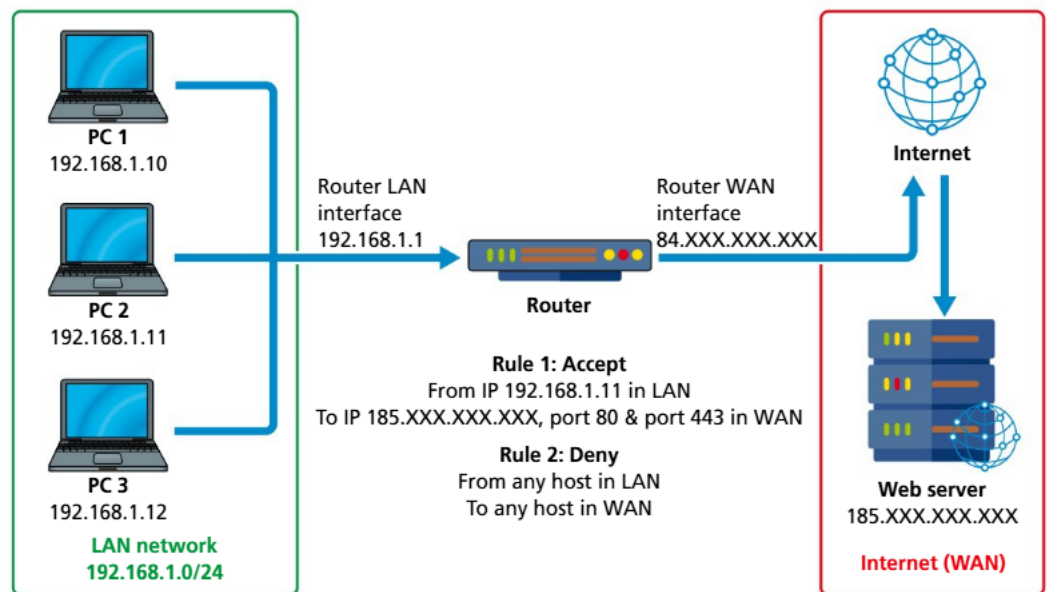
SYLLABUS CONTENT

By the end of this chapter, you should be able to:

- ▶ A2.4.1 Discuss the effectiveness of firewalls for protecting a network
- ▶ A2.4.2 Describe common network vulnerabilities (HL)
- ▶ A2.4.3 Describe common network countermeasures (HL)
- ▶ A2.4.4 Describe the process of encryption and digital certificates

A2.4.1 Firewalls

Firewalls obviously play a key role in helping protect computer networks, but what exactly do they do?



■ Firewall in action

A firewall inspects the IP packets that pass through it, and filters the incoming and outgoing traffic, based on a set of predefined security rules. In the illustration above, Rule 1 is allowing traffic from PC2 on the internal LAN to access any external IP address that begins with 185.??? for destination ports 80 or 443. Rule 2 then denies all other internal requests. Each individual packet is checked against the rules until a rule is encountered that allows or denies passage through the firewall.

Some firewalls are also capable of applying stateful inspection, where the state of active connections is monitored and can be used to determine passage through the firewall, for instance once a connection has been allowed to be established, packets are allowed to continue back and forth through the connection until it closes.

The rules comprise a set of allow lists and deny lists. These lists are IP addresses or ranges of IP addresses, and optionally can also specify application port numbers.

Using the rule lists, firewalls control access to the internal network from external sources, and can also be used to control which external destinations are accessible from internal sources. Additionally, firewalls log traffic requests, which can be useful for detecting and responding to suspicious activity or for identifying a potential breach.

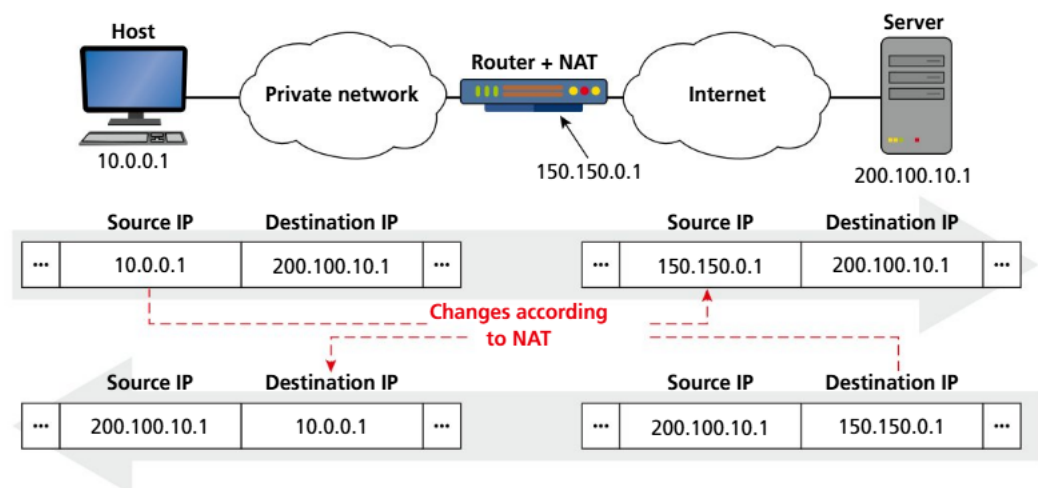
Firewalls are just one tool for securing a network, and are not perfect for protecting against all types of attacks. For instance, firewalls are less effective against threats from within the network such as a malicious insider, or if an external threat has gained local presence on the network via compromised WiFi. Some threats can be quite sophisticated and can mimic legitimate traffic. Finally, a firewall is only as good as its configuration. If a network manager doesn't keep the deny lists updated and the firewall firmware up to date, then vulnerabilities can be exposed.

Common mistake

Over-reliance on firewalls

Don't assume that a firewall alone is sufficient to protect a computer network. Firewalls are gatekeepers for ranges of addresses and ports. Most firewalls do not inspect the content of the data beyond the source and destination address and port, meaning malicious data can easily still be let through if destined for a legitimate location.

Network address translation



Network address translation in action

You were previously introduced to the concept of network address translation in Section A2.3.1. Beyond the convenience that network address translation has provided through allowing the internet to continue to function despite having exhausted IPv4 addresses years ago, it has also had an additional benefit with respect to network security.

Since NAT modifies the IP address information in packet headers, the process helps to hide internal IP addresses on a network, which makes it more difficult for an external attacker to reach internal systems. Although it is not technically a security feature by design, it does provide an additional layer of difficulty that attackers must overcome, as the internal structure of your network and IP addresses are not directly exposed. In this sense, it could be described as a case of security by obscurity.

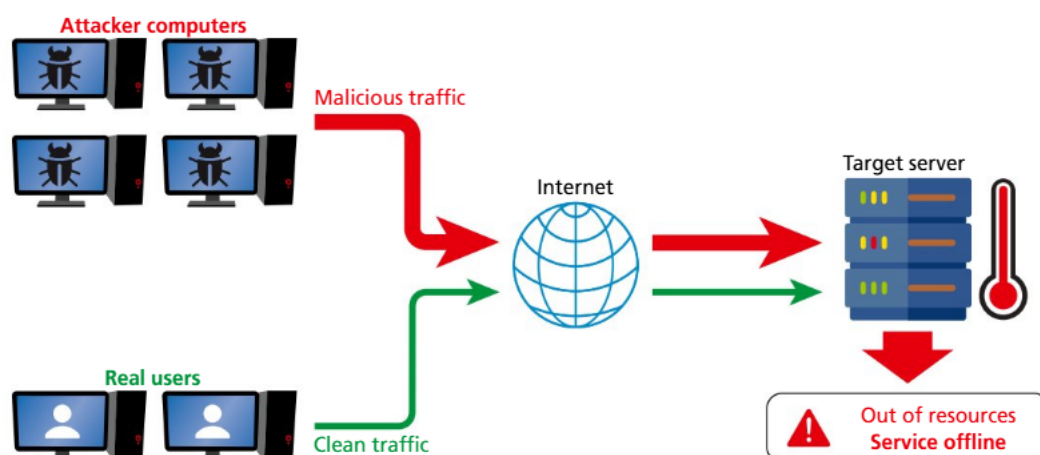
A2.4.2 Common network vulnerabilities (HL)

In this section, we briefly consider a range of common attack vectors and vulnerabilities within computer networks. In the subsequent section, we review strategies to mitigate these risks.

● Linking question

What are the similarities and differences between network security and database security? (A3)

■ Distributed denial of service (DDoS)



A distributed denial of service (DDoS) attack is one in which multiple systems under the control of the malicious actor generate a massive number of network requests, with the goal of overwhelming the target system. This can cause the target to slow down, so that it becomes unusable or unavailable for legitimate users.

It is important to point out that the computers being used for the attack are typically not even devices directly owned or operated by the attacker. It is more likely that they are devices that have fallen victim to having unwittingly installed malware on their system that the attacker can then use to contribute towards the overall DDoS attack.

■ Insecure network protocols

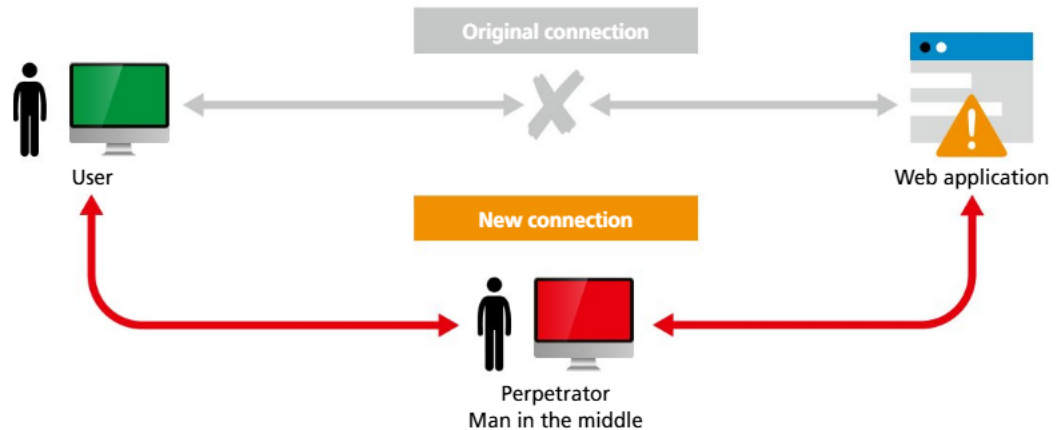
There are many older or poorly designed network protocols still in widespread use that do not include modern security features. These older protocols allow attackers to intercept, or even alter and manipulate, the content of data in transit on the internet.

Examples of these older protocols are HTTP, FTP and Telnet, all of which transmit their data in the clear without any form of encryption. The modern alternatives are HTTPS, SFTP and SSH.

■ Malware

“Malware”, or “malicious software”, is the overarching term for a classic range of attacks that have had various names applied to them over the years. If you’ve come across terms such as “computer viruses”, “worms”, “trojan horses” and “ransomware”, these are all various forms of malware. They are software that is designed to harm or exploit vulnerable devices and to seek out and spread to other vulnerable devices. They can lead to data loss, data theft and loss of full control of the device. Malware can sometimes sit idle for extended periods of time until triggered to activate (such as in the case of a DDoS client).

■ Man-in-the-middle (MitM)



A man-in-the-middle attack (now also sometimes referred to as an on-path attack) is where an attacker eavesdrops on communications between two systems.

This can result in data breaches, or data alteration, without the knowledge of the original parties involved.

■ Phishing

Phishing is when a victim receives, and responds to, a seemingly legitimate message (such as an email) that has the aim of deceiving the victim into providing sensitive information, such as login credentials or bank account details. This can result in unauthorized access to other systems, identity theft and financial crimes.

■ SQL injection

An SQL injection is where an attacker exploits vulnerabilities in the way an application or website has been designed to work with its database. It relies upon the programmers not applying sufficient validation checks to the user inputs, so that the application can make unintended changes to the underlying database. This can result in unauthorized viewing of user lists (and their passwords), deletion of data or granting administrative control of the database to the attacker.

To understand how the SQL injection attack works, consider the impact the following Python would have to the SQL statement if executed on a database:

```
first_name = "Robert'); DROP TABLE Students; --"
family_name = "Doe"
sql = "INSERT INTO Students VALUES ('" + first_name + "', '"
+ family_name + "')";
```

The SQL string now actually contains three different SQL commands:

- 1 INSERT INTO Students VALUES ('Robert');
- 2 DROP TABLE Students;
- 3 -, 'Doe');

Never trust inputs from users or feed them straight into your database without verifying and validating them first!

■ Cross site scripting (XSS)

Cross site scripting attacks operate on similar principles to the SQL injection attack. Instead of using insufficient validation to send rogue database instructions, a cross site scripting attacker injects their own code to run client-side on pages viewed by other users. This can bypass access controls, deface websites or redirect users to malicious sites.

Any website that loads and executes JavaScript that is being hosted by a third party (a very common practice for many JavaScript libraries) makes itself potentially vulnerable to this scenario, as the website is effectively inviting the third party to run their own code on the website.

■ Unpatched software

Unpatched software refers to software that has not had all published security updates applied to it. Once security updates are released by software vendors, it is particularly important to apply them as soon as possible as not only do they help address and correct a vulnerability, they also advertise the existence of the vulnerability, making devices lacking the update even more vulnerable to attack.

■ Weak authentication

Systems that have poorly designed authentication systems make themselves vulnerable to attack by design. Examples of this might be websites that do not properly hash and salt their passwords, or that store their security keys in insecure folders. Multifactor authentication systems, and the use of tools such as OAuth, are industry standard for a reason.

Developers should be discouraged from rolling out their own authentication system for anything other than as a learning tool. Use and rely on the excellent high-quality authentication systems that have been built upon the tears of those who have gone before you.

■ Zero-day exploits

A zero-day exploit is when an attacker takes advantage of a previously unknown weakness in software or hardware, before its maintainers have had the opportunity to create a patch to fix the vulnerability. These types of exploits are especially dangerous because, by their very nature, there is no known defence against them.

Perhaps the best-known historical example of a zero-day exploit was Stuxnet. It was malware that was used to break into Iran's uranium-enrichment centrifuges in 2006, and is suspected to have been created by the USA's National Security Agency (NSA).

A2.4.3 Common network countermeasures (HL)



Common mistake

Network security is more than having anti-malware software and using encryption

Many times, when students are asked how to protect a computer or data on a network, they simply revert to saying “install anti-malware software” or “use encryption”. These are lazy answers that do not capture the nuance of the approaches outlined in this section. Network security requires a complex, multi-layered approach to protect against attacks. Be sure to understand the different strategies outlined and be prepared to discuss them in assessments.



Top tip!

Match the threat to the countermeasure

One method of ensuring you master a more detailed understanding of the complexities of securing a computer network is to review the various threats discussed in the previous section, and match each to the appropriate countermeasure discussed in this section. For instance, multifactor authentication can help counter weak authentication issues.

■ Content security policies

The content security policy header is settings that can be placed into the HTTP header to define which sources are permitted to load content on to a particular website. Properly set, they can help guard against XSS attacks and similar injection exploits.

For more information, search online for Mozilla’s content-security-policy documentation (<https://developer.mozilla.org/en-US/docs/Web/HTTP/Headers/Content-Security-Policy>).

■ Complex password policies

Password policies are continually a vexed and contentious issue.

Some in the industry continue to argue for password rules that enforce the creation of strong, hard-to-brute-force passwords that require minimum lengths and a mix of letters, numbers and special characters.

Others in the industry argue that best practice should be to help users cope with what is described as “password overload”. Security experts such as Troy Hunt argue best practice should entail:

- only using passwords where they are really needed
- using technical solutions to reduce the burden on users
- allowing users to securely record and store their passwords
- only asking users to change their passwords on suspicion of compromise
- allowing users to reset passwords easily, quickly and cheaply.

For more on modern understanding of password policies, search online for Troy Hunt’s excellent article “Passwords Evolved: Authentication Guidance for the Modern Era” (www.troyhunt.com/passwords-evolved-authentication-guidance-for-the-modern-era).

Major companies such as Microsoft and Google are now promoting the use of passwordless authentication as much as possible. As Google states: “Developers and users both hate passwords: they give a poor user experience, they add conversion friction, and they create security liability for both users and developers.”

Passkeys are being promoted as a safer and easier alternative. This can take one of a variety of forms, including:

- biometric sensor login (fingerprint or face recognition)
- PIN
- pattern.

Read more about passkeys by searching online for Google Developer’s “Passwordless login with passkeys” (<https://developers.google.com/identity/passkeys>).

If you do use a website that still requires a password, be concerned if it imposes a maximum length on the password you set. If a password is properly hashed, it won’t matter how long it is, so if you are told by a system that the password has a maximum length, chances are it means they are storing it in unhashed clear text! Run, run far away, and definitely do not reuse a “real” password on that service! (The author has been known to use variations of “thisPasswordIsNotSecure” on such websites 😊)

■ Distributed denial of service (DDoS) mitigation tools

To help guard a server against a DDoS attack, most firewalls and web servers have settings available that help absorb or deflect traffic overloads. They do this by rate limiting (allowing a maximum number of requests in a given time-period from any individual source), traffic analysis and also subscribing to cloud-based DDoS protection services, such as those provided by the major cloud-hosting companies.

■ Email filtering solutions

One of the most common attack vectors used for the spread of malicious software remains email. To this end, having modern and up-to-date email-filtering systems to scan incoming emails for malicious attachments, phishing attempts and general marketing spam can go a long way to protect end-users from email-based threats.

■ Encrypted protocols

To protect against man-in-the-middle and other similar attacks is trivially easy now. Simply do not use unencrypted protocols such as HTTP, FTP or Telnet. Only use systems that accommodate the secure, encrypted modern versions of HTTPS, SFTP or SSH.

■ Input validation

A maxim in Computer Science is never to trust user inputs: always validate them!

Use a range of input-validation tools, such as presence check, length check, type check and format check to validate that the structure of the incoming data is in the style and format expected. For instance, your software shouldn’t be accepting kilobytes of data when the entry box is for a person’s name or date of birth!

A few simple checks on the input data being received from the user, prior to accepting and processing it, goes a long way to protecting systems from harmful data.

An example of where lack of input validation caused global security concerns was the Heartbleed vulnerability that was hidden within the important OpenSSL cryptography library, which is a widely used implementation of the transport layer security protocol.

For a simplified look at how the attack worked, search online for “1354: Heartbleed Explanation” (www.explainxkcd.com/wiki/index.php/1354:_Heartbleed_Explanation).

■ Intrusion detection systems (IDS) and intrusion prevention systems (IPS)

Intrusion detection and prevention systems are specialized software tools that actively monitor network traffic and the broader state of your system. Their goal is not just to detect potential threats, but proactively to block them before they succeed in compromising your systems.

The intrusion *detection* system focuses on monitoring network and system traffic to spot suspicious activities and potential threats. It sends alerts to network administrators when it identifies such activity. The IDS is a passive system, meaning that it does not interfere with the flow of traffic on the system.

The intrusion *prevention* system is an active system. When it detects a threat, it takes proactive measures to prevent the threat from gaining access to, or harming, the network. It can block traffic, drop malicious packets, close connections and more.

In modern cybersecurity solutions, the IDS and IPS roles are often bundled together into a comprehensive security product, allowing for both detection of threats and active measures to prevent those threats from causing harm. These modern tools have taken on the name of “endpoint protection” to describe their role. Endpoint-protection software typically also bundles other security functions, such as antivirus, anti-malware, firewalls and others, with the goal of integration being to provide a robust protection regime.

Most operating systems come with their own basic endpoint-protection utilities installed (in Windows it is known as Defender), but there are many commercial operators in this space. CrowdStrike is one such operator, which became famous for all the wrong reasons when, on 19 July 2024, it sent out a faulty update for its Falcon Sensor product that resulted in over 8 million Windows computers getting stuck at the infamous Blue Screen of Death (BSOD), affecting critical systems for airlines, banks, hospitals, supermarkets and many other organizations around the globe.

■ Multifactor authentication (MFA)



■ One-time passcode

Multifactor authentication systems are an easy-to-implement measure that require more than one method of authentication by users to verify their identity.

The commonly stated goal of introducing multiple factors is to require two of the three means by which a user can prove their identity:

- Something they know (e.g. a password).
- Something they have (e.g. their phone – by way of receiving or generating a code).
- Something they are (e.g. a biometric, such as fingerprint or facial recognition).

As a consumer, you should enable two-factor authentication on as many services as possible and use an app such as Authy.

● Top tip!

Adding one-time codes to your projects

As a beginner software developer, there are libraries available that make it very easy to incorporate one-time codes that are compatible with all the major authentication apps. For Python, search online for the PyOPT library (<https://github.com/pyauth/pyotp>).

■ Secure socket layer (SSL) certificate and transport layer security (TLS) certificate

SSL and TLS certificates allow consumers to authenticate the developer or organization that authored a software product. Developers and companies can use these certificates to digitally sign the project as their work, thereby giving trust and confidence to the consumer. These digital certificates are discussed in more detail in Section A2.4.4.

■ Update software

Keeping your software up to date by regularly applying all the latest security patches is an important step for mitigating the risk of falling victim to a security vulnerability.

■ Virtual private networks (VPN)

As previously discussed, VPNs provide a secure means to exchange data with a remote office as if your device was connected directly to the private network. Companies regularly require their employees to make use of VPN technologies whenever they are working remotely away from the office.

■ Testing and training

Regular testing of your security measures, including “white hat” penetration testing and vulnerability assessments, forms a critical part of ensuring systems remain secure. Testing helps ensure your systems still work as you expect, and that no new weaknesses have emerged. As part of the testing, there should also be regular training for staff within an organization. Many security breaches occur through human error, so training employees in security best practices on an ongoing basis is also a crucial part of any security regime.

■ Wireless security measures

Wireless networks pose a unique vulnerability to any computer network and require special security measures to identify and mitigate risks. One commonly used measure is to restrict access to the network to each device’s unique media access control (MAC) address. The MAC address is a unique 48-bit address assigned to each network card by the manufacturer, so these can be used to populate an allow-list of devices permitted to connect to the network.

■ Secured backups

When was the last time you backed up your files?! For data that matters to you, take responsibility for its care and protection. Don’t just rely on cloud services either, as they are known to fail occasionally. Purchase a spare portable drive to keep in the back of your sock drawer, and copy your data to it at least once a month.

Top tip!

Data synchronization tool

For students confident with using console commands, `rsync` is the best tool for copying files for backups. Search online for “manpagez: man pages & more man `rsync(1)`” for the documentation (www.manpagez.com/man/1/rsync/).

If you are less confident with the console, search online for “Free File Sync”, a free, open-source file synchronization tool (<https://freefilesync.org>).

Finally, if your backups resemble a collection of portable drives, you should consider the security of your backups! An attacker doesn't need to penetrate your network if they can just pinch a USB drive!

Top tip!

File-encryption tool

Consider encrypting your backups with a tool such as `gnupg` (<https://gnupg.org/download>): just don't forget the password!

■ To encrypt:

```
gpg --output encrypted_backup.enc --symmetric --cipher-algo AES256 mybackup.zip
```

■ To decrypt:

```
gpg --output mybackup.zip --decrypt encrypted_backup.enc
```

Linking question

Do networks and databases use the same form of **encryption** algorithms? (A3)

◆ Encryption:

the conversion of information or data into a mathematically secure format that cannot be easily understood by unauthorized people.

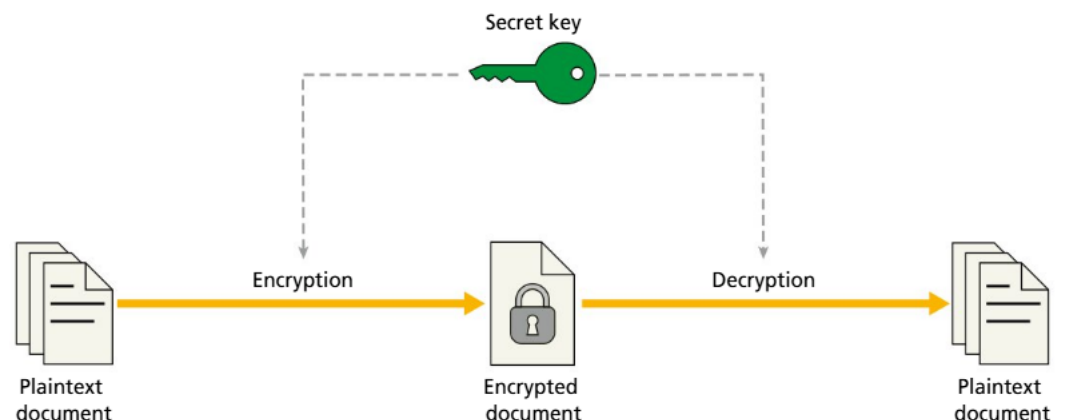
A2.4.4 The process of encryption and digital certificates

Common mistake

Thinking digital signatures and digital certificates are the same thing

Ensure you understand the nuance of the distinction between a digital signature and a digital certificate. They are not the same thing.

Symmetric encryption



◆ **Encryption key:** a string of characters or numbers used by an encryption algorithm to encode or decode data. It is the values that are input into the mathematical functions responsible for scrambling or descrambling the data.

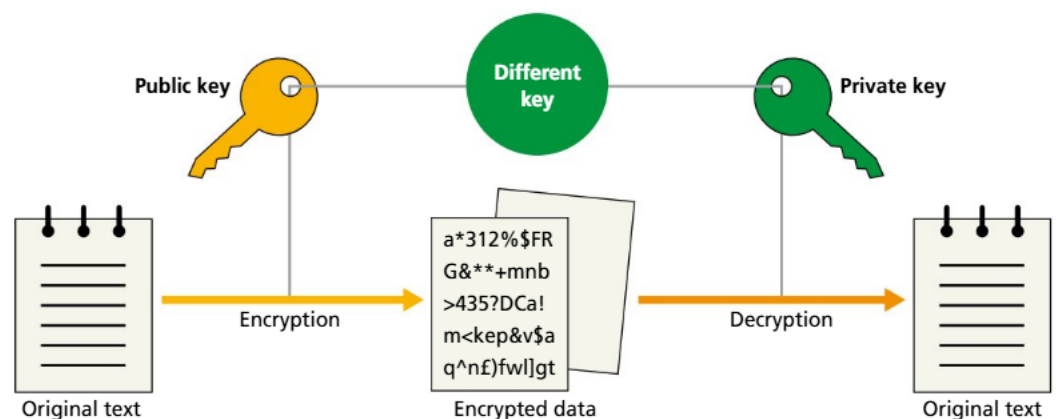
Symmetric encryption is where the same **key** is used for both encrypting and decrypting the data. This means the key must be shared between both the sender and receiver in a secure manner.

Symmetric encryption is generally faster and less computationally intensive than its asymmetric equivalent, so it makes sense to use it for encrypting large files.

When physical distance separates the sender and receiver, it can be difficult to share the encryption key in such a manner as to be protected against eavesdroppers. There are two common solutions to this problem:

- Use asymmetric encryption to establish secure communications. Use this asymmetric method to exchange and agree on a symmetric key, and then switch communications to the faster, more efficient symmetric approach.
- Use a mathematically secure method to exchange keys, such as Diffie Hellman (covered later in this section).

■ Asymmetric encryption



Asymmetric encryption is when the security key used to encrypt data is different from the key used to decrypt it. If it can work successfully, there are significant benefits to be gained, as it means two parties wanting to communicate in encrypted form do not have to meet privately to exchange an agreed-upon encryption key, as is required for symmetric encryption. Instead, the encryption key can be published publicly, and any person wishing to send a message can use it to encrypt the data, so that only the recipient and key-holder can decrypt it to read.

For example: If Alice wants to send a secure message to Bob, she encrypts it using Bob's public key. Only Bob can decrypt this message with his private key.

How does it work? How is it algorithmically possible to use one cipher key to encode the data and a different key to decode it? The answer is mathematics! The following walkthrough, inspired by an example by Henry J. Schmale, is simplified and loosely based on the RSA (Rivest–Shamir–Adleman) encryption algorithm.

● Key information

The mathematical walkthrough is provided to demonstrate how such algorithms are possible, as an exercise of intellectual interest. Understanding the mathematics is beyond the scope of the syllabus. You will not be examined on this procedure or asked to do these calculations in the exams.

Step 1: Generate your public and private key

Start by selecting two prime numbers. These are generally very large: up to 2048 bits is typical. We will use 61 and 53 for our walkthrough.

$$p = 61$$

$$q = 53$$

Find n , the product of the two primes, and $\lambda(n)$ (known as “Carmichael’s totient function”).

To follow the maths, all you need to appreciate is that it is the least common multiple of $(p - 1)(q - 1)$.

$$n = pq = 61 \times 53 = 3233$$

and

$$\lambda(n) = \text{lcm}[(p - 1)(q - 1)] = \text{lcm}(60, 52) = 780$$

Select an integer, e , that is a prime number, less than $\lambda(n)$, and not a factor of $\lambda(n)$. A common choice is $2^{16} + 1$, being 65537.

$$e = 17$$

Solve d , where $(d \times e) \bmod \lambda(n) = 1$. This is known as the “modular multiplicative inverse”.

That is, $(17 d) \bmod (780) = 1$. One valid answer in this case is $d = 413$.

The public key will be the two numbers $n = 3233$, $e = 17$, and the private key will be the two numbers $n = 3233$, $d = 413$.

Step 2: Encrypt your secret message

The cipher text, c , for a secret message, m , is given by the following equation:

$$c(m) = m^e \bmod n$$

So, if the secret message is A , which is ASCII 65:

$$c(65) = 65^{17} \bmod 3233 = 2790$$

Step 3: Decrypt your secret message

The message, m , for cipher text c is given by the following equation:

$$m(c) = c^d \bmod n$$

So, when given cipher text 2790, this would be decrypted as follows:

$$m(2790) = 2790^{413} \bmod 3233 = 65^{17}$$

The security of the algorithm rests in the size of the prime numbers and that factorizing a number into its two constituent primes is still not possible except by brute force of trying every possible set of values. The original prime number factors are required to be able to calculate $\lambda(n)$. In this case, it would be trivial to find the factors of 3233, but finding the factors of a 4096-bit number would take hundreds or thousands of years.

One area of active computing research that may threaten the security of RSA-encrypted data is quantum computing. Shor’s algorithm, specifically, is a quantum algorithm for the factorization of integers that could work very quickly and render modern cryptography obsolete.

■ Role of digital certificates

Digital certificates are used as a way of certifying identity on the internet. For HTTPS traffic, they are issued by a mutually trusted third party known as a “certificate authority (CA)”.

When a digital certificate is presented in a network transaction, it helps the recipient verify that the public key belongs to the sender and not an imposter. Digital certificates form a key part of the network of trust on the internet. It’s not enough to know that your communication

with `yourbank.com` is encrypted, if any other person can pose as the legitimate web server for `yourbank.com`. You want to know the web server you are logging in to is the one you want to share your secrets with.

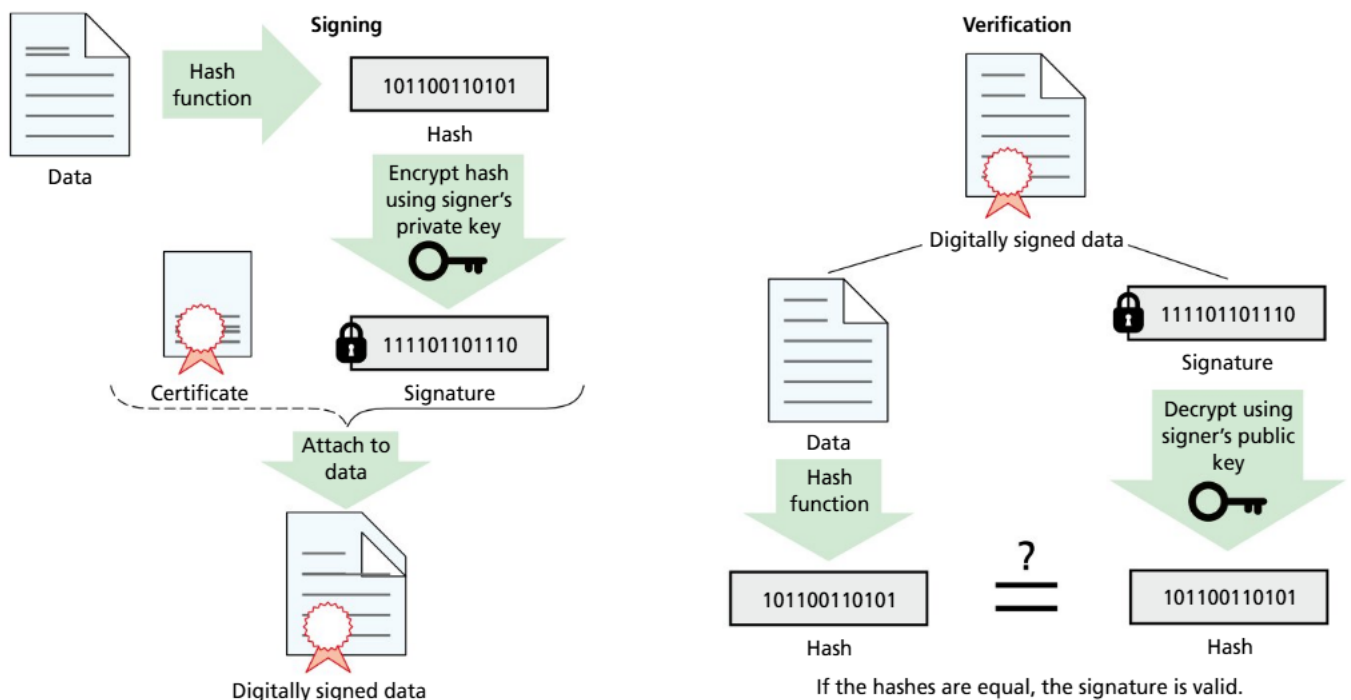
Obtaining a certificate

The process starts by `yourbank.com` requesting that a certificate authority (that is mutually trusted by both parties) issues a certificate that can be used to prove they are, indeed, the legitimate server for `yourbank.com`. The bank will generate a public and private key using a process similar to that previously discussed with the RSA algorithm. The private key is kept secure on the bank's server, while the public key is sent to the certificate authority with the request for certification.

Using the public key provided in the certification request, the CA will verify the identity and legitimacy of the bank. This process varies, and is dependent on the process applied by the certificate authority. In the early days of the internet, it required sending photo ID proof, business ownership certificates and other legal documents to the CA. Modern practice has simplified this significantly to encourage broader adoption of HTTPS, so now certificate authorities such as `letsencrypt.org` offer the service for free and without complex paperwork.

Once the CA is satisfied with the verification, it will use its private key to sign the public key of the bank or other website. This signing process involves creating a hash of the certificate that is then encrypted using the CA's private key. This encrypted hash becomes the electronic signature. It can't be produced without the CA's private key, but it can be validated through using the CA's public key.

Using digital certificates



■ Signing with a digital certificate

Once issued, the digital certificate can be attached to all communications from the bank or other organization, and used to cryptographically sign the communication. This provides a mechanism of allowing recipients to algorithmically verify the authenticity of the origin of the communication.

Digital certificates can also be self-generated to create public and private keys that can be used for secure login to network services, such as via SSH (Secure Shell). In this case, your public key is uploaded to the remote machine you want to authenticate yourself to later and you keep the private key secure on your own local machine. The private key is equivalent to your password. When you want to log in via SSH, the server will generate a challenge to you that is encrypted with the public key. You, as the sole person in possession of the private key, are therefore the only person able to decrypt the challenge and thereby prove your identity.

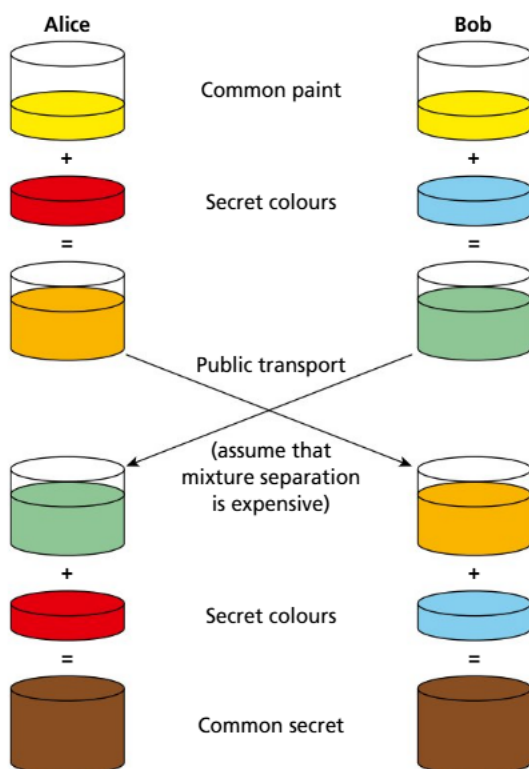
Given the crucial importance of cryptographic keys in the modern interconnected economy, proper storage of keys is essential. If anyone gains unauthorized access to an organization's private keys, they can act as an imposter of that organization in all electronic transactions for which that key has been set up to be used. The creation, distribution, usage, storage and eventual retirement and deletion of encryption keys is therefore a key task of any effective IT infrastructure.

Digital certificates are a core part of blockchain technologies such as cryptocurrencies. All transactions on a blockchain are both signed and hashed. Senders use their private keys to digitally sign transactions as part of the process of validating themselves as the owner of the cryptocurrency that they are spending. This signature serves to authenticate the identity of the sender and ensure the non-repudiation of the transaction. The sender cannot later deny having made the transaction.

ACTIVITY



Thinking skills: Spot the forgery at Coding Quest in the 2022 challenge (<https://codingquest.io/problem/5>).



■ Diffie Hellman key exchange

Given that asymmetric encryption algorithms require significantly more processing than symmetric algorithms, it is ideal if lengthy communication sessions can occur using symmetric encryption. The obvious problem that occurs then is to find a secure method of exchanging and agreeing on the symmetric key. While asymmetric encryption could be used to do this, it is slower and more processing intensive. An alternative approach that is commonly used is known as the "Diffie Hellman key exchange".

The analogy used to describe Diffie Hellman is the difficulty of unmixing colours of paint. While we may have a vague idea of which colours might be used to constitute brown, it is almost impossible to unmix the input colours perfectly.

From an algorithmic perspective, Diffie Hellman relies on the same principles of asymmetric encryption with respect to prime numbers and the mathematical difficulty of determining the prime factors of a number once run through a modulus operation.

This is a mathematical walkthrough of the Diffie Hellman algorithm:

Step 1

Agree (in public) on a base and a modulus. The modulus must be a prime number.

Person 1

$base = 109, modulus = 811$

Person 2

$base = 109, modulus = 811$

Step 2

Pick a secret number as your exponent. Calculate the base, raised it to the secret, and then put it through the modulus.

$secret = 197$

$109^{197} \% 811 = 679$

$secret = 312$

$109^{312} \% 811 = 337$

Step 3

Exchange the result of step 2 with your friend.

$received = 337$

$received = 679$

Step 4

Calculate your common secret by taking the received number, raising it to your original secret and then running the result through the modulus.

$337^{197} \% 811 = 215$

$679^{312} \% 811 = 215$

Key information

Video walkthrough

Search for the YouTube video "Diffie-Hellman Key Exchange: How to Share a Secret" by Spanning Tree.

REVIEW QUESTIONS

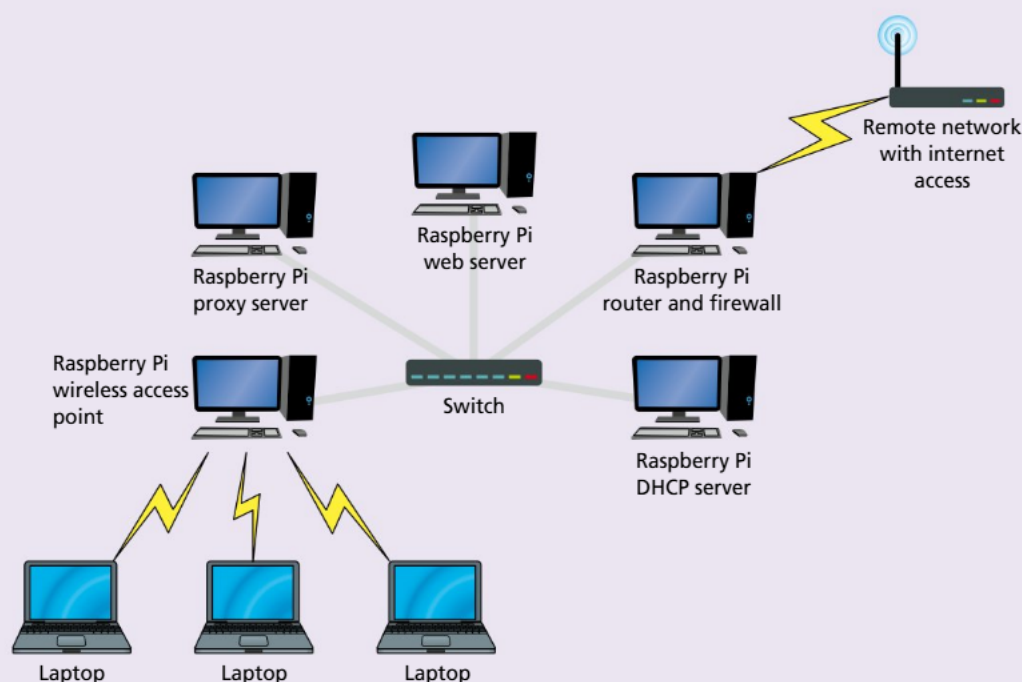
- 1 What is a primary function of a firewall in a network security context?
 - a Monitoring network performance
 - b Inspecting and filtering traffic based on set rules
 - c Increasing the speed of network traffic
 - d Providing physical security to network devices
- 2 Which attack involves overwhelming a service with excessive requests in order to make it unavailable?
 - a Phishing
 - b SQL injection
 - c DDoS
 - d MitM
- 3 What type of attack intercepts and possibly alters the communication between two parties who believe they are directly communicating with each other?
 - a DDoS
 - b MitM
 - c Phishing
 - d XSS
- 4 Which vulnerability arises when software has not been updated to address known security issues?
 - a Insecure network protocols
 - b Unpatched software
 - c Weak authentication
 - d Malware

- 5 Which network countermeasure is specifically designed to prevent unauthorized access by verifying users through multiple methods?
 - a Multifactor authentication (MFA)
 - b Intrusion detection system (IDS)
 - c Virtual private network (VPN)
 - d Email filtering
- 6 Which countermeasure can help protect a network against interception and unauthorized access to data in transit?
 - a Multifactor authentication (MFA)
 - b Intrusion prevention system (IPS)
 - c Transport layer security (TLS) certificate
 - d Virtual private networks (VPN)
- 7 Which security practice involves verifying the legitimacy of a website's connection to ensure it is secure?
 - a Applying content security policies
 - b Updating software regularly
 - c Using secure socket layer (SSL) certificates
 - d Email filtering
- 8 Which type of cryptography uses the same key for encryption and decryption?
 - a Symmetric
 - b Asymmetric
 - c Both A and B
 - d Neither A nor B
- 9 Describe how network address translation (NAT) contributes to enhancing network security.
- 10 Describe what is meant by "zero-day exploit".
- 11 Describe SQL injection and how it can affect a database-driven website.
- 12 Describe how weak authentication can pose a risk to network security.
- 13 Describe the purpose of using intrusion detection systems (IDS) and how they function.
- 14 Describe how DDoS mitigation tools work to protect a network.
- 15 Describe the role of digital certificates in establishing secure network connections.



ACTIVITY

Thinking skills: Design a network with Raspberry Pi



■ Raspberry Pi network structure

Create a small network using Raspberry Pi to simulate real-world network scenarios, including client-server and peer-to-peer architectures, routing and basic network security implementations.

If you are completing this task as a class, split into small groups, allocating one Raspberry Pi per group. Assign each Raspberry Pi one of the following roles:

- Firewall and router
- Proxy server
- Web server
- DHCP server
- Wireless access point server
- DNS server (optional)

The following instructions suggest utility programs and configuration files that would be useful for achieving your objectives but lack sufficient particularity to complete the instruction without further research by your team. The information provided is mostly to help you know you are on the right track when searching for guides online.

Materials needed:

- Several Raspberry Pi computers (at least one per group; ideally three or four per group for diverse roles)
- SD cards with Raspberry Pi OS installed
- Ethernet cables
- Network switch (or a router with multiple Ethernet ports)
- Monitors, keyboards and mice for interfacing with the Raspberry Pi computers
- Internet connection (for downloading packages and updates)

Initial set-up for all Raspberry Pi computers:

- Each Raspberry Pi needs the Raspberry Pi OS installed and updated. You can download the latest version from the official Raspberry Pi website and write it to an SD card. Connect the Raspberry Pi to a keyboard, monitor and mouse and power up. The default login credentials are username `pi` and password `raspberrypi`. When configuring the Raspberry Pi computers, enable `SSH` and `VNC` to ease remote access later.
- You may need occasionally to temporarily connect the Raspberry Pi computers directly to an internet connection, rather than to your custom network, for the purposes of installing updates and other programs.

For the Raspberry Pi designated as router and firewall:

- 1 Ensure the Raspberry Pi is obtaining an IP address from the DHCP server. This Pi will need a static IP (so other Pi computers can find it to request internet traffic), so the DHCP server will need the MAC address of this Pi so as to assign it the same IP address each time. To view the IP address from the console, use the command `ip a`.
- 2 To set a static IP directly (without the DHCP server), edit the file `/etc/dhcpd.conf` as follows:

```
interface eth0
static ip_address=192.168.1.1/24
static routers=192.168.1.254
static domain_name_servers=192.168.1.254
```

- 3 Enable IP forwarding by editing `/etc/sysctl.conf`.
- 4 Set up the WiFi connection to your school WiFi network (or other network providing outbound internet access) by running `sudo raspi-config`.
- 5 Configure NAT and the firewall with `nftables`. The main configuration file will be `/etc/nftables.conf`. A basic template configuration may resemble:

```
table inet filter {
    chain input {
        type filter hook input priority 0;
        # Accept any localhost traffic
        iif lo accept
        # Accept traffic already established
        ct state established,related accept
        # Enable HTTP and HTTPS
        tcp dport { http, https } accept
        # Drop everything else
        counter drop
    }
}
```

- 6 To apply change to rules, run this command: `sudo nft -f /etc/nftables.conf`.
- 7 Reboot the Pi for settings to take effect, and then test.

For the Raspberry Pi designated as wireless access point:

- 1 Install `hostapd` as your access point host software.
- 2 The main configuration file will be `/etc/hostapd/hostapd.conf`. Suggested settings include:

```
interface=wlan0
bridge=br0
driver=nl80211
ssid=PiNet
hw_mode=g
channel=7
wmm_enabled=0
macaddr_acl=0
```

```
auth_algs=1
ignore_broadcast_ssid=0
wpa=2
wpa_passphrase=YourPasswordHere
wpa_key_mgmt=WPA-PSK
wpa_pairwise=TKIP
rsn_pairwise=CCMP
```

- 3 Configure `hostapd` to use that configuration file by editing `/etc/default/hostapd`.
- 4 Prevent the wireless network card from being managed by the default network manager by changing `/etc/dhcpd.conf` and specifying `denyinterfaces wlan0`.

For the Raspberry Pi designated as DHCP server:

- 1 Install `isc-dhcp-server` as a DHCP server.
- 2 The main configuration files will be `/etc/dhcp/dhcpd.conf` and `/etc/default/isc-dhcp-server`.
- 3 An example template for `/etc/dhcp/dhcpd.conf` is:

```
subnet 192.168.1.0 netmask 255.255.255.0 {
    range 192.168.1.10 192.168.1.100;
    option domain-name-servers 192.168.1.1;
    option routers 192.168.1.1;
    default-lease-time 600;
    max-lease-time 7200;
}
```

- 4 Ensure the DHCP server is configured to start on boot.
- 5 Check the logs for errors by using `cat /var/log/syslog | grep dhcp`.

For the Raspberry Pi designated as web server:

- 1 Ensure the Raspberry Pi is obtaining an IP address from the DHCP server. This Pi will need a static IP (so other Pi computers can find it to request a webpage), so the DHCP server will need the MAC address of this Pi so as to assign it the same IP address each time. To view the IP address from the console, use the command `ip a`.
- 2 Install `nginx` as the web server software, and set it to start automatically on boot.
- 3 Create a web page at `/var/www/html`.
- 4 The main configuration files for `nginx` can be found in `/etc/nginx`.
- 5 If `nginx` fails to start, you can check for configuration errors by using `sudo nginx -t`.
- 6 An example HTML file for `/var/www/html/index.html` might resemble:

```
<!DOCTYPE html>
<html lang="en">
<head>
    <meta charset="UTF-8">
    <title>Welcome to Pi Web Server</title>
</head>
<body>
    <h1>Hello, Raspberry Pi Network!</h1>
    <p>This is a simple web page served from the Raspberry Pi web server.</p>
</body>
</html>
```

For the Raspberry Pi designated as proxy server:

- 1 Ensure the Raspberry Pi is obtaining an IP address from the DHCP server. This Pi will need a static IP (so other Pi computers can find it to request a webpage), so the DHCP server will need the MAC address of this Pi so as to assign it the same IP address each time. To view the IP address from the console, use the command `ip a`.
- 2 Install `squid` proxy server software, and set it to start automatically on boot.
- 3 The main configuration file is at `/etc/squid/squid.conf`.

- 4 In addition to acting as a web cache, because all traffic passes through a proxy it can also be used as a content filter, for example you could block requests for given URLs. This is similar, though functionally different, from the firewall approach. Investigate how to use squid to block certain websites.

For the Raspberry Pi designated as DNS server:

- 1 Ensure the Raspberry Pi is obtaining an IP address from the DHCP server. This Pi will need a static IP (so other Pi computers can find it to request a webpage), so the DHCP server will need the MAC address of this Pi so as to assign it the same IP address each time. To view the IP address from the console, use the command `ip a`.
- 2 When the configuration for the DNS server is complete, the DHCP server will need updating so clients know to look to this computer for DNS resolution queries.
- 3 Install `bind9` as your DNS server software, and ensure that it is set to start on boot.
- 4 The main configuration files will be in the folder `/etc/bind`.
- 5 To check the configuration for errors, go to `sudo named-checkconf` and `sudo named-checkzone myhome.local /etc/bind/zones/db.myhome.local`.
- 6 An example ZONE file `/etc/bind/db.myhome.local` to create your own DNS domain (that would work on your internal network) might resemble:

```
$TTL      604800
@          IN      SOA      ns1.myhome.local.      admin.myhome.local. (
                        3              ; Serial
                        604800         ; Refresh
                        86400          ; Retry
                        2419200        ; Expire
                        604800 )       ; Negative Cache TTL
;
@          IN      NS       ns1.myhome.local.
ns1        IN      A        192.168.1.1
@          IN      A        192.168.1.1
www        IN      A        192.168.1.2
```

ATL alignment links

Research skills:

- Finding and selecting appropriate software and hardware resources.
- Understanding documentation to configure each Raspberry Pi for its specific role.
- Comparing solutions and different methods or options to determine what will work best for project requirements.

Thinking skills:

- Problem-solving the numerous challenges that come in setting up a new network, such as IP address conflicts, firewall rules and ensuring all devices can communicate effectively.
- Critical thinking to evaluate the effectiveness of each network component and troubleshoot issues.
- Decision-making the design of the network.

Social skills:

- Collaboration by working together in a group to divide tasks, share findings and support each other.
- Conflict resolution of different opinions on how to configure the network or solve problems.
- Responsibility-sharing and relying on the contributions of others.

Communication skills:

- Technical writing to document the network set-up and configuration steps.
- Oral communication to explain complex concepts such as DNS or DHCP with peers or teachers.

Self-management skills:

- Planning and organization to manage a large, cumulative task.
- Adaptability to deal with unexpected issues.
- Self-motivation and initiative to go beyond a basic set-up, such as implementing additional network services or security measures.

EXAM PRACTICE QUESTIONS

1 Small business network set-up

A small business has recently expanded and needs to update its network infrastructure to support more employees and provide secure, efficient access to resources.

- a Identify one piece of network hardware that would connect multiple computers within the office to create a local area network (LAN). [1]
- b The business has chosen a star topology for its network. Describe one advantage of using a star topology in a small business environment. [2]
- c The company uses both HTTP and HTTPS protocols on its internal websites. Describe one key difference between HTTP and HTTPS. [2]
- d In the context of the TCP/IP model, describe the function of the transport layer. [3]
- e The network architecture includes both client-server and peer-to-peer set-ups. Compare these two types of network architecture in terms of resource management. [3]
- f The business plans to use IPv6. Identify one advantage of using IPv6 over IPv4 in a growing business. [1]
- g The business is choosing between using fibre-optic cables and wireless transmission for connecting different departments. Discuss two factors that should be considered. [2]
- h Describe how routers in the network use dynamic routing to manage data traffic. [3]
- i The company is concerned about network security. Describe one common network vulnerability and a corresponding countermeasure. [3]
- j Data encryption is crucial for the company's operations. Describe the difference between symmetric and asymmetric encryption in the context of data security. [3]
- k Outline how digital certificates contribute to network security. [2]

2 University campus network

A large university campus network needs to support thousands of users in a dynamic, high-demand environment.

- a State one network device essential for connecting campus buildings spread over a wide area. [1]
- b The university uses a hybrid network topology. Describe one benefit of using a hybrid topology in a university setting. [2]
- c DNS servers are critical in a university network. Describe the role of DNS in network operations. [2]
- d Describe how the internet layer of the TCP/IP model facilitates connectivity between different buildings on campus. [3]
- e The campus uses both IPv4 and IPv6 addressing. Outline these two types of IP addressing in terms of address availability. [2]
- f Evaluate the use of twisted-pair cables vs fibre-optic cables for data transmission within academic buildings. Consider factors such as cost and speed. [3]
- g Explain how packet switching enables efficient data-traffic management on a busy university network. [3]
- h Describe one network security measure that should be implemented to protect students' grades and other sensitive data from alteration by the students themselves, while still allowing them access to the campus network. [2]
- i The university employs digital signatures for document verification. Describe how digital signatures enhance security. [3]
- j Outline the use of certificate authorities in the university's network-security framework. [2]

3 Airline network infrastructure

An airline operates a comprehensive network infrastructure to manage its public-facing website for customer bookings, co-ordinate hundreds of staff across multiple locations and handle customer interactions through self-check-in kiosks.

- a i Identify one network device that is essential for connecting the airline's global offices to its central database. [1]
- ii Outline one key network-security technique that should be used. [2]
- b The airline's network uses a mesh topology for critical systems. Outline one advantage of using a mesh topology for such applications. [2]
- c HTTPS is mandated for the airline's booking website. Describe the role of SSL/TLS in HTTPS. [2]
- d Describe the role of the application layer in the TCP/IP model, particularly in processing online bookings. [3]
- e The network includes a mix of wired and wireless technologies. Compare these two technologies in terms of reliability and security, specifically for use in high-traffic customer areas such as airports. [3]
- f Describe the importance of using IPv6 for the airline's network, focusing on its ability to handle numerous devices and security features. [1]
- g Describe how network routers ensure data packets find the optimal path across complex networks. [3]
- h Data security is a major concern for the airline, especially with customer data. Describe one potential network threat and a preventative measure that can be implemented. [3]
- i The airline uses VLANs to segment network traffic. Explain how VLANs enhance network security and efficiency. [3]
- j Outline the significance of using a reverse proxy in managing the high traffic on the airline's booking website. [2]
- k Describe the role of firewalls in protecting the network infrastructure of the airline, especially in scenarios involving customer data and payment transactions. [2]

